Lesson Notes

MODULE 2 | LESSON 1

RESAMPLING EFFICIENT FRONTIERS

Reading Time	90 minutes
Prior Knowledge	Mathematics of classical portfolio theory and certain statistical knowledge
Keywords	Resampling, resampling efficient frontiers, Michaud's optimization, Morninstar's EnCorr, bootstrap techniques.

This lesson will describe the methodology of resampling the efficient frontier to obtain more robust estimates that do not suffer the same drawbacks as when practitioners rely on the simple optimization technique described in Lesson 4 of Module 1.

1. General Mean-Variance Optimization Problem

In Lesson 4 of Module 1, we learned several things about the efficient frontier. However, most of what we did was find the best portfolio for us given a required rate of return, which is the very definition of efficient frontier. There is no arguing that this is relevant. Still, if we recall the way we got to the formulation of the efficient frontier starting from an individual's utility function, i.e.,

$$U(w() = U(\bar{w}) + U'(\bar{w})(w - \bar{w}) + \frac{1}{2}U''(\bar{w})(w - \bar{w})^2 + R_3,$$

we concluded that individuals have a preference for expected wealth, $\bar{w}$, ($U(\cdot)$ is an increasing function of its argument), and also, individuals who are risk-averse will have an aversion to risk ($U''(\cdot)$ has negative sign). This leads us to write the following maximization program:

$$\begin{aligned} &\max_{w} w^T r - \frac{1}{2} w^T \Sigma w \\ &\text{s.t. } w^t \iota = 1, \end{aligned}$$

where the symbols have the same meaning as before. This is the standard program, but we could add additional constraints like maximum exposure to any single stock, minimum exposure, etc. We are also aware that, even if agents are risk-averse, it is quite possible that not everyone has the same risk-aversion to volatility. We could modify the QP above to read as follows:

$$\max_{w} w^T r - \frac{\alpha}{2} w^T \Sigma w$$
$$\text{s.t. } w^t \iota = 1,$$

where $\alpha \geq 0$ measures the intensity of risk-aversion.$\alpha = 0$ would describe an individual who has no consideration for risk,$\alpha = 1$ would describe our representative risk-averse agent,$\alpha = 2$ an investor who feels a much stronger dislike for risk than someone with $\alpha = 1,$ and so on.

Using the same portfolio of seven stocks as before and the same reference period, we find that for a traditional risk-averse individual $(\alpha = 1)$:

```
optimal weights:
0.7652182 1.7812181 2.0609607 -0.3036158
-1.2216889 0.6933467 -2.7754390
return achieved on portfolio:
1.175
volatility of optimal portfolio:
1.0166
```

On the other hand, if the individual had much less appetite for risk (say the individual is near retirement and does not have enough years to make up for losses in their portfolio) and applies a larger penalty to risk, say $\alpha = 2,$ then we find:

```
optimal weights:
0.44033322 0.92022890 1.09255015 -0.04042926
-0.58977574 0.34565857 -1.16856583
return achieved on portfolio:
0.6811266
volatility of optimal portfolio:
0.5410759
```

This is all interesting and one could toy with it by changing the value of the penalty factor α and investigate what happens to the solution.

In fact, it would be a great exercise for students who would like to master working with matrices (an important mathematical tool, in general) to solve this problem analytically instead of relying on numeric methods. The solution for the weights turns out to be:

$$w_{opt} = \frac{\Sigma^{-1}\iota}{\iota^T \Sigma^{-1} \iota} + \frac{1}{\alpha}\left(\Sigma^{-1} - \frac{\Sigma^{-1}\iota\iota^T\Sigma^{-1}}{\iota'\Sigma^{-1}\iota}\right) r.$$

Immediately, we notice that if $\alpha \to \infty,$ we get the weights for the minimum variance portfolio! That is what we would expect: If someone cannot stand any bit more variance than necessary, the global

minimum variance portfolio is the only optimal portfolio for them. Hence, the solution we found makes sense.

2. Parameter Estimation and Issues

The reason why we considered the analytical solution is not just for the sake of doing some mathematics. The real motivation rests with the fact that the formula shows how the optimal portfolio (and similarly the efficient frontier) depends heavily on inputs like the covariance matrix, Σ, and the vector of expected return, r.

In practice, the parameters necessary to derive the efficient frontier are not known with certainty and, anyway, are bound to change over time. As we rely on estimates based on historical returns to estimate these parameters, the solution of the optimization problem is subject to instability in the sense that even small changes in the inputs could produce large differences in the optimal portfolios that we computed. In other words, noisy inputs lead to unstable portfolios.

In particular, it so happens that when a few assets have much higher average returns than others, a portfolio will tend to concentrate on those assets and this leads to portfolios that are not well-diversified. The issue is not so much the type of solution that we achieved and which is correct based on the inputs, but the fact that if the economic scenario changes, we will not reap the benefits of diversification.

Also, noisy estimates, changing a lot over even relatively short periods of time, lead to portfolios that require frequent re-balancing and which can be expensive to implement.

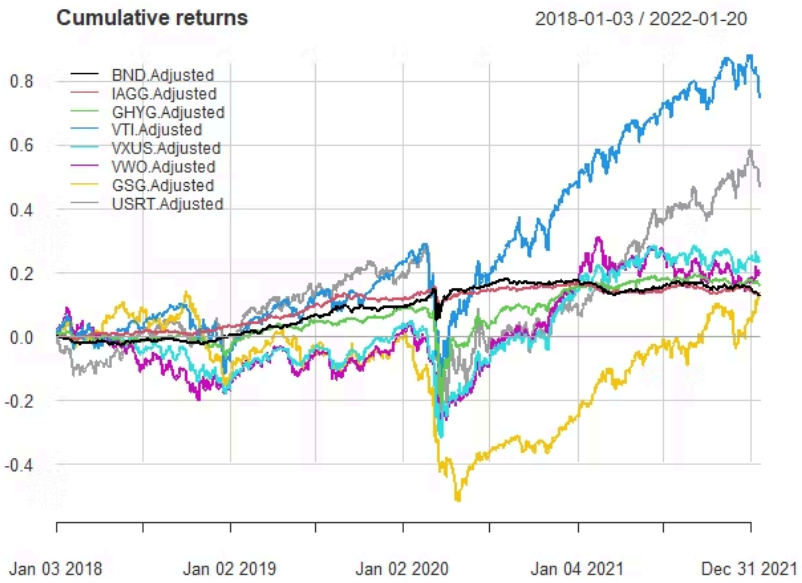
The COVID-19 crisis illustrates the instability we refer to. In order to make the example most useful, we will not use just U.S. equities but will instead look for different classes of assets to try being encompassing of a large variety of assets. The assets/classes of investments we will use are the following:

- the Vanguard Total Stock Market Index (VTI);
- the Vanguard FTSE Emerging Markets Index Fund ETF Shares (VWO);
- the iShares Global High Yield Corporate Bond (GHYG);
- the Vanguard Total Bond Market Index Fund ETF Shares (BND);
- the iShares Core International Aggregate Bond ETF (IAGG)
- the Vanguard Total International Stock ETF (VXUS)
- the iShares S&P GSCI Commodity-Indexed Trust (GSG)
- the iShares Core U.S. REIT ETF (USRT)

We consider the period between 2018 and the current date so that we have roughly two years before and after the start of COVID-19 crisis.

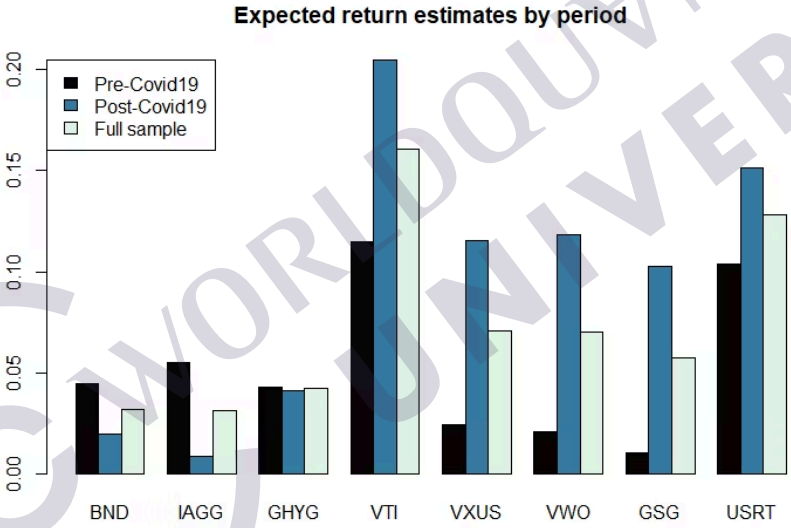
The plot of the cumulative returns is very clear in showing how much the behavior of the assets has changed over the two periods before and after COVID 19.

Figure 1: Cumulative Returns for Asset Prices for the Period Indicated



Let's look at the expected returns and the volatility over the two sub-periods and the entire sample, which is done in Figure 2.

Figure 2: Expected Returns Pre-COVID 19, Post-COVID 19 (Full Sample Period)



The bar chart for volatility shows a similar pattern of variation over the two subperiods (Figure 3). And it is likely that the correlations have changed as well to a certain extent.

Figure 3: Volatility Pre-COVID 19, Post-COVID 19, and over the Full Sample Period

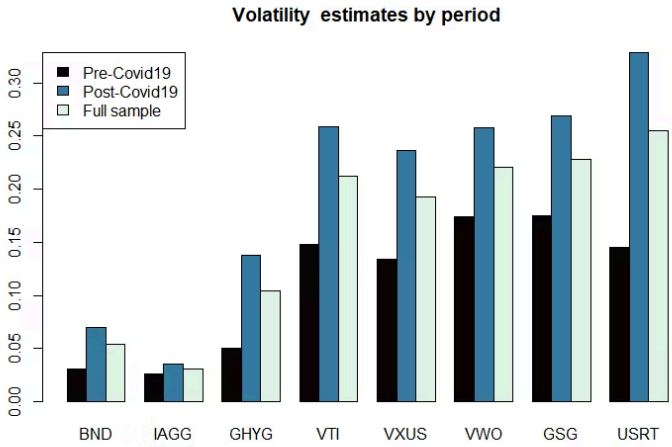


Figure 4 shows how the efficient frontiers are different among the different periods. This clearly leads to different weights of the individual assets in a portfolio and may produce high concentrations on fewer assets and force substantial and costly re-balancing. Also, notice that for during the COVID-19 period, the tangency portfolio is found much further to the right due to the much higher volatilities and the higher expected returns.

Figure 4: Efficient Frontiers Pre-COVID 19, Post-COVID 19, and for the Full Sample Period (The dots on each curve represent the tangency portfolio)

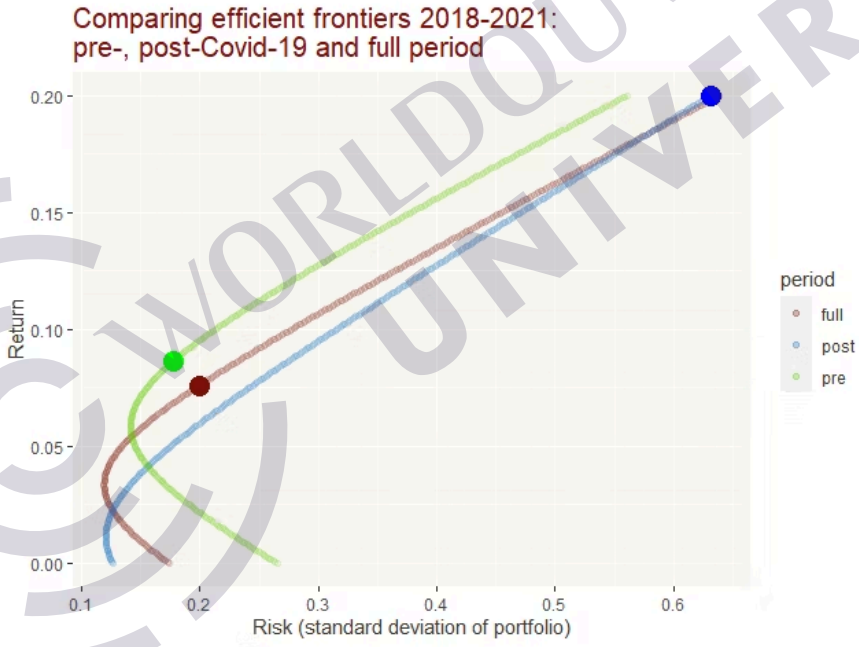
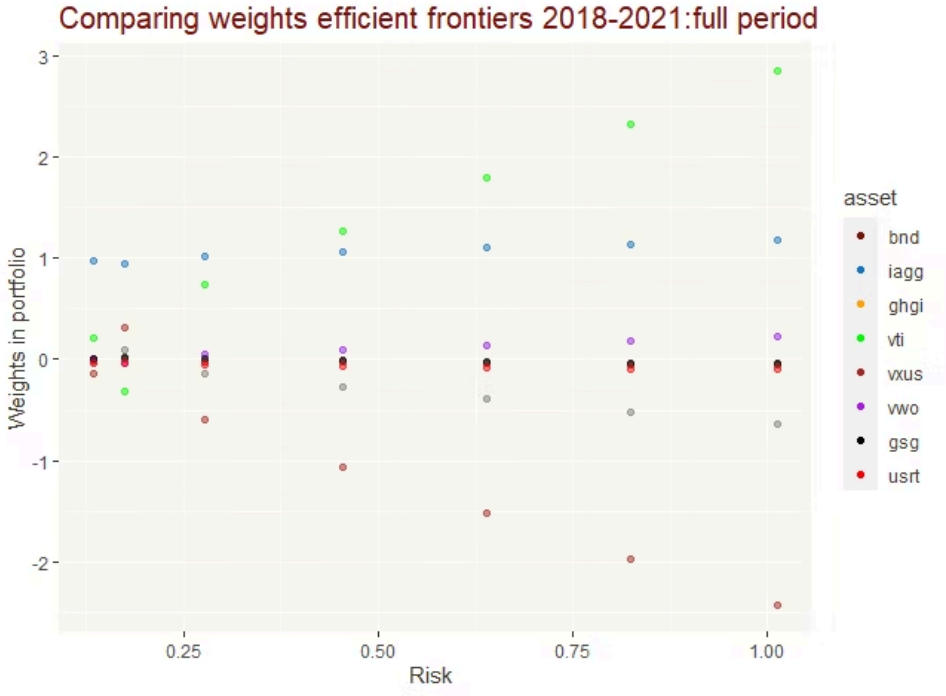


Figure 5 looks at the weights for the optimal portfolios based on specific amount of risks. This plot shows two important features:

- the portfolio tends to consist of just a few of the 8 assets (VTI, IAGG, VXUS) and as the desired return increases (and thus does the risk/volatility of the optimal portfolio), and
- a long position in VTI and a short position in VXUS tend to dominate all other weights more and more.

We looked at the results for the full period, but those for the other two subperiods are similar.

Figure 5: Weights of Assets for Portfolios on the Efficient Frontier



There is a relatively plausible explanation for the fact that the weights of some of the assets become bigger and bigger in proportion to those of the other assets. As the algorithm tends to maximize Sharpe's ratio and short selling is allowed, it achieves this by aggressively selling short stocks with the lowest estimated mean return and buying stocks with the highest estimated mean returns. Adding constraints to our optimization problem would alleviate the issue to some extent, but it would not change the fact that we are relying on estimated parameters and that even small differences in the estimates can produce large effects on the results.

3. Resampling the Efficient Frontiers

One solution to the problem of concentrated positions is to add constraints to the optimization problems. For instance, we could impose that, under no circumstances, the weights of all or some assets may exceed a certain % of the total portfolio. Similarly, we could impose restriction about the amount of short-selling that is allowed or even demand that there be no short-selling at all. Overall, these expedients will not allow any asset to become dominant or to be used as a source to generate funds (via short-selling).

Nevertheless, because the efficient frontier and the optimal portfolio rely heavily on the estimated mean returns, $\hat{\mu}$, and variance-covariance matrix, $\hat{\Sigma}$ it is quite possible that our estimates are off by quite a bit vs. the true (and unknown!) value of these parameters, μ and Σ, respectively. We should reflect on the fact that we only see one sample out of an entire population of possible returns for our assets during the reference period. In particular, mean returns are very prone to estimation errors.

R. Michaud in 1998 began to pioneer the use of resampling techniques dealing with the problem of global asset allocations. In fact, Michaud has built a business around resampling (and as a matter of fact, the procedure was patented in 1999, U.S. Patent #6,003,018). His techniques have a name as well, **Michaud optimization or resampled efficiency.** Still, he is not the only one to have thought about this since Morningstar also has its own resampling technique, which is incorporated in their **EnCorr** software (Moringstar EnCorr User Guide). The two methodologies use different

approaches and interested readers can find a comparison between the two techniques in the following paper: Morningstar vs. Michaud Optimization.

Michaud's procedure produces multiple sets of statistically equivalent (i.e. coming from a common underlying distribution) risk-return estimates around the original estimates. When the terminology ""statistically equivalent"" is used, it means that they all come from the same population/distribution. These sets of statistically equivalent returns allow us to compute many efficient frontiers, each optimized with respect to the particular sample we got from our assumed distribution for the inputs. In fact, each frontier will be different from the others, some maybe even very different from one another. The Michaud method calculates its efficient frontier by averaging all the frontiers that were produced using simulations. His method works like this:

Step 1. Given the time series of returns, estimate the expected returns of the m assets, which we call $\hat{\mu}$ and the variance-covariance matrix $\widehat{\sum}$ These can be estimated using MLE or any other statistical techniques (some may produce better estimates than others).

Step 2. With $\hat{\mu}$ and $\widehat{\sum}$ estimated earlier, we can determine the minimum variance portfolio and the maximum return portfolio using the suitable formulae

$$w = argmin_{w \in C} w^T \widehat{\Sigma} w$$

where C is the set of constraints for our problem, and

$$w = argmax_{w \in C} w^T \hat{\mu}$$

We call the return of the portfolio associated with these two systems of weights L and H, respectively. Note that if we set limits to the portfolio exposure to each assets, the maximum return will be finite

Step 3. Divide the range $L, H]$ in say $N-1$ subintervals (the reason for $N-1$ is that we want to have N points). This yields the following expected returns:

$$L \equiv E[r_{GMV}], r_2 \equiv E[r_{GMV}] + \delta, \ldots, r_N = E[r_{GMV}] + (N-1)\delta = H$$

with

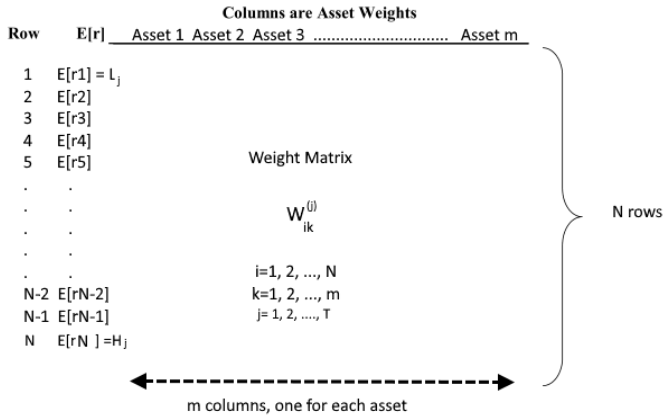
$$\delta = \frac{H - L}{N - 1}.$$

Step 4. This is were the simulations begin. We make an assumption that the returns are distributed as a multivariate normal with mean $\hat{\mu}$ and covariance matrix $\widehat{\sum}$.We extract T random samples from this distribution. (Note that because we are sampling from a multivariate normal with covariance $\widehat{\sum}$, the returns we are sampling are not uncorrelated in general and reproduce the dependency that we see in the real markets). In other words, we are generating T samples of m returns, one for each of our assets from a population $N(\hat{\mu}, \widehat{\Sigma})$. .

Step 5. For each of the generated T samples, we estimate a new mean expected return, say $\tilde{\mu}_j$ and a new covariance matrix, say $\tilde{Sigma}_j$ where $j = 1, 2, \ldots, T$. Then, we use these estimates to compute an efficient frontier, say $\mathbf{EF}_j$ using the N return points in the interval $[L, H]$. At this point,

for each j, we should have a matrix of weights $W_{ik}^{(j)}$ like in the picture immediately below. Note that at this stage it is better to replace L and H with L_j and H_j, but the process works exactly as described in Step 2.

Figure 6: Resampling Weights using Michaud's Efficient Resample Technique

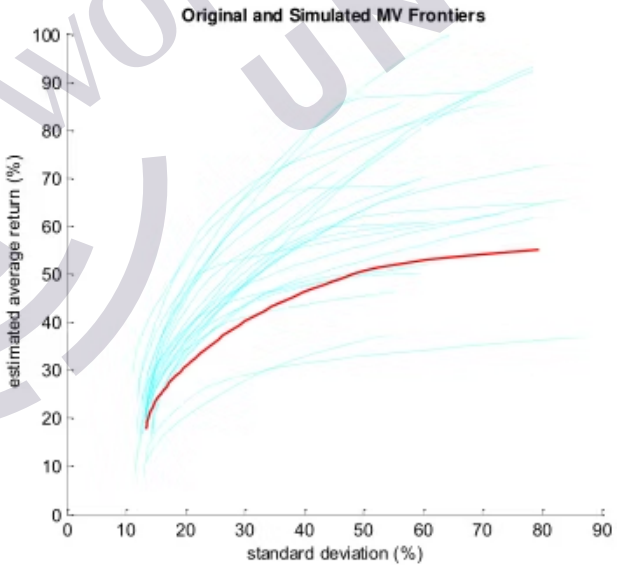


Step 6. We have now T matrices of weights as described in the picture above. That means that for every entry W_{ik} we have T values, $W_{ik}^{(j)}, j = 1, 2, \ldots, T$, one value for each simulation. Then we average each element of the matrix over to get $W_{ik}^{*} = \frac{1}{T} \sum_{j=1}^{T} W_{ik}^{(j)}$.

Step 7. With the new weights W_{ik}^{*} we can redraw the efficient frontier by using the original means, $\hat{\mu}$ and variance-covariance matrix $\hat{\Sigma}$. This yields the resampled efficient frontier.

What we get after all this looks like the picture immediately in Figure 7.

Figure 7: The Original Efficient Frontier (red) and the Resampled Frontiers (light blue)



Source: Michaud, Richard & Michaud, Robert. (2007). Estimation Error and Portfolio Optimization: A Resampling Solution. Journal of Investment Management. Vol. 6. pp. 8 - 28. Image reproduced with permission by the authors.

Resampling efficient frontiers with a large number of portfolios and implementing it properly to guarantee convergence requires a very heavy computational burden. At the same time, this is one of those problems that can benefit from parallel programming.

Once we determine an efficient frontier using Michaud's resampling methodology, if we notice that the efficient frontier obtained from the original data is sufficiently close (say in a 95% confidence

level region), we can use our result with a certain degree of confidence. However, if we notice that the efficient frontier is quite far apart from Michaud's, we cannot simply attribute the differences to statistical factors, and we must investigate the problem further.

Also, we have proposed a multivariate normal as our underlying distribution, but we could have used a multivariate t distribution or other distributions, in principle whenever we have a reason to decide that one family of distributions is better than the others.

Overall, estimating ρ and Σ using sample means is not considered the best possible approach and other estimating procedures are possible. One such method is **shrinkage estimation**. Let's say that ρ_i is the mean for asset i(out of m assets), and we define $\tilde{\rho} = \frac{1}{m}\sum_{i=1}^{m}\rho_i$, i.e., it is the mean of means (Statisticians refer to it as the "grand mean").

The shrinkage estimator for the mean is simple:

$$\tilde{\rho}_i = \gamma\rho_i + (1-\gamma)\tilde{\rho}.$$

This procedure is implemented to bring the means closer to each other. One may wonder why. The rationale is to reduce the variance of the estimator. Can it be done at no cost? No, there is a price to pay and the tradeoff is that using a shrinkage methodology introduces bias (unless the true means for the returns happen to be all the same, an unlikely event).γ is the shrinkage parameter, and it is a number between 0 and 1. If $\gamma = 0$ all $\tilde{\rho}_i = \tilde{\rho}$, the grand mean and the variance is minimal. However, in that case, the bias is large. Shrinkage estimators, we stress, are only one approach and do not always produce dramatic improvements, but they do work in many cases.

There is a shrinkage estimator for the covariance matrix as well, but that is more complex than for the case of the mean. One approach suggests to write the covariance matrix as $\Sigma = M\Lambda M^T$ where M is the matrix of normalized eigenvectors and Λ is the matrix of the eigenvalues of Σ. Note, that this can be done because of specific theorems in linear algebra.

Then, one "shrinks" the eigenvalues of the matrix Λ. For instance:

$$\tilde{\lambda}_i = n\lambda_i/\alpha_i$$

and

$$\alpha_i = n - p + 2\lambda_i\sum_{j\neq i}\frac{1}{\lambda_j - \lambda_i}$$

But, this approach (known as the *Stein estimator*) does not work all the time because it can produce negative eigenvalues, which should not happen when a matrix is positive definite like a covariance matrix. There are others, but this is not the place for a sophisticated investigation of the topic. Students interested in this topic should consult B. Efron and T. Hastie's Computer-Aided Statistical Inference, Chapter 7, Section 1, pp. 91-94.

In general, the means are the critical input. The variance/covariance matrix is important too, but while even small changes in the estimated means can have a large impact in the determination of the optimal portfolio weights, this is not so for the variance/covariance matrix.

The relevant takeaway is that when working with small samples, a shrinkage estimator may help improve the results obtained from resampling procedures.

Michaud's technique has been tested by practitioners and academics over the years. For instance, in 2003, Markowitz and Usmen compared it to the Bayesian portfolio optimization method, and while Markowitz and Usmen expected the Bayesian strategy to do better than the resampling approach, surprisingly, Michaud's strategy did better. But not everyone agrees that this conclusion holds all the time. Students who are interested in this topic should read A Theoretical Foundation of Portfolio Resampling.

Morningstar's EnCorr uses resampling as well. However, it uses a different approach. Morningstar's algorithm divides the risk/volatility spectrum spanned by the Markowitz efficient frontier into equal lengths of bins of standard deviations. The minimum bin starts at the minimum variance portfolio and the maximum bin ends at the maximum return portfolio. EFs are simulated using a volatility within its bin. The resulting Morningstar efficient frontier is computed as the average of the portfolios in each bin and displayed as the risk and return of the average portfolio.

Note: Michaud's portfolio resampling methodologies is a patented procedure. This means nobody can use or distribute the algorithm without licensing it or obtaining permission. This is important for a quantitative analyst as much as for a portfolio manager since the use of the methodology without licensing could produce monetary fines for the professional or the firm. It would also reflect badly in their business relations with clients and may be in breach of the fiduciary duty to clients.

Students interested in understanding more in depth the nuances of portfolio resampling, and not just Michaud's, are encouraged to read the paper by Frahm listed among the references.

4. Accuracy of Estimates of the Sharpe Ratio for the Tangency Portfolio

Assume that we want to find the tangency portfolio that maximizes the Sharpe ratio. This is something that we know how to do very well by now using our returns and the estimated parameters (mean returns and covariance matrix). Clearly, as this is just one sample of returns over the reference period out of many possible, we would like to check whether the Sharpe ratio of the portfolio we find is an over- or an under-estimation of the actual one.

The bootstrap technique is very useful to answer this type of question. In this case, we treat the sample we have as if it were not a sample, but the actual population of stock returns. Consequently, the mean returns and covariance matrix are no longer estimates but true parameters. In other words, we make the sample available to be our population and the Sharpe ratio for the tangency portfolio from this sample population becomes the true Sharpe ratio.

Now we employ a resampling scheme via bootstrap and not by simulating new data. The process is rather simple and works like this:

Step 1. Draw M samples from the data (resamples).

Step 2. Decide on the resample size $n < N$, with N the number of returns for each asset in our portfolio. Deciding what n to use is a practitioner decision. Sometimes, one tries the experiment using different values of n.

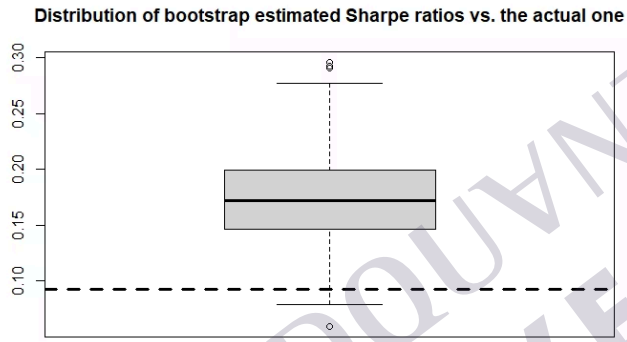
Step 3. For each resample, we estimate a new vector of mean returns and a covariance matrix. With these, we can compute the optimal (Sharpe) portfolio. This will give us M optimal portfolios. Overall, we will have M systems of weights and M Sharpe ratios, one for each of the bootstrap samples.

Step 4. Take the mean of the weights for the M optimal portfolios.

Step 5. The re-sampled efficient frontier is then found by multiplying the average weights from Step 4 by the actual returns and the risk by using the covariance formula for the portfolio and the same weights from Step 4.

The results of our simulations are shown in Figure 8. The bootstrap experiment shows that our process to identify the efficient frontier tends to overestimate the actual one, that is things look better than they really are. The data used for the simulation is obtained from the pre-covid period without short sale restrictions.

Figure 8: Distribution of Sharpe Ratios vs. Actual Ones using Bootstrap Resampling



The overestimation of the efficient frontier and the Sharpe ratio produces higher confidence in the process than it probably should. Can the issue be alleviated or reduced? If we introduced short-sale restrictions, one could see that the bootstrap estimates fall much closer to the actual Sharpe ratio.

Statistical remark about bootstrapping with time series: This seems relatively simple and, to a certain extent, it is. However, we should stress the fact that we are bootstrapping multivariate data and, in particular, time series. When resampling multivariate time series data, the dependencies within the observation vectors must be preserved. In other words, we cannot just adopt a re-sampling scheme with replacement because that would destroy dependency. So how do we do it? Thankfully, it is not too complicated. We assumed earlier that we had samples of N data (i.e., we know the returns for all our assets during a period of N days, months, or whatever the reference period is. Then, we identify our sample rows with a number: 1, 2, ..., N. When we take the M resamples, we resample the number of the row and then we take the returns for all assets along that particular row.

5. Conclusion

We derived the general mean-variance optimization solution and noticed that it relies heavily on the estimated mean vector and covariance matrix for assets' returns over the reference period. When these returns are non-stationary, the algorithm produces solutions that can be very volatile and concentrated. To mitigate these issues, we introduced Michaud's optimization, or resampled

efficiency, and learned how to derive a better efficient frontier, which takes into consideration that mean returns and covariance matrices are not parameters but simply estimates. In the last section, using bootstrap, we saw that the traditional method of deriving optimal portfolios overestimates the efficient frontier and thus predicts better results than one would actually achieve.

We stress here, once again, that Michaud's methods cannot be used without license or permission because it is patented. The next lesson will broaden the problem of optimal portfolio selection by adding skewness, kurtosis, and liquidity to the problem.

References

- Frahm, G. "A Theoretical Foundation of Portfolio Resampling." *Theory and Decision*, 2014, pp. 1–30, http://dx.doi.org/10.2139/ssrn.2085586.
- Michaud, Richard O., and Robert O. Michaud. "Estimation Error and Portfolio Optimization: A Resampling Solution." *The Journal of Investment Management*, vol. 6, no. 1, 2007, pp. 1–8.

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Lesson Notes

MODULE 2 | LESSON 2

BEYOND MEAN-VARIANCE OPTIMIZATION

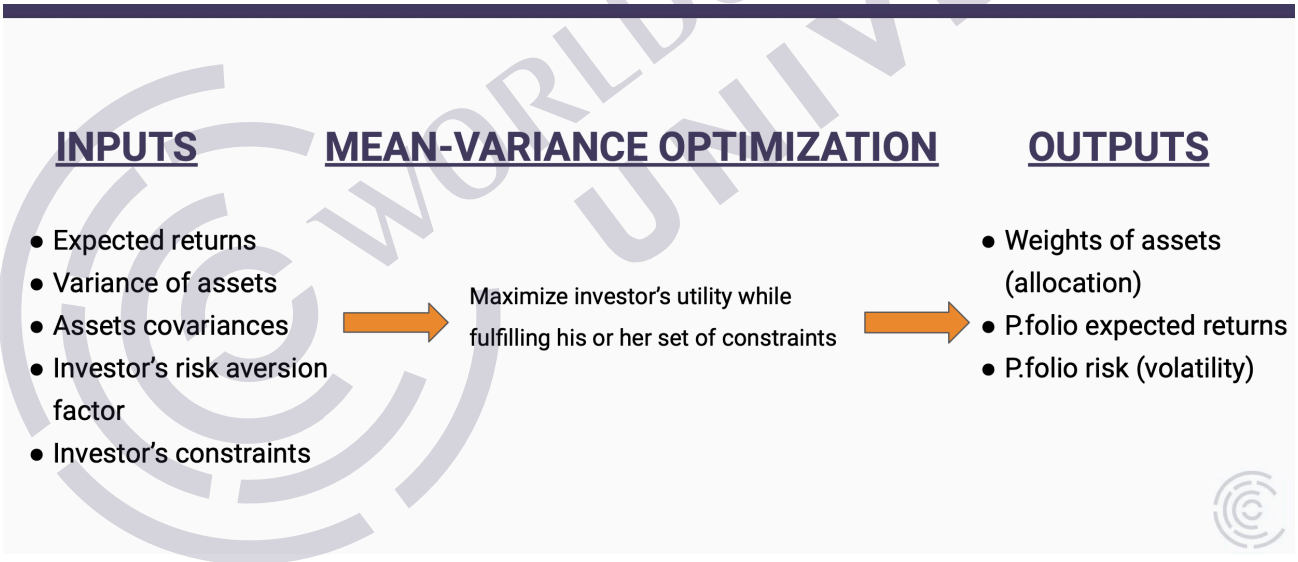
Reading Time	150 minutes
Prior Knowledge	A sound understanding of the previous lesson, and an ability to manipulate matrices and using symbolic calculations (scipy library in Python)
Keywords	Coskewness, Cokurtosis, Kronecker products, General optimization (beyond quadratic programming), Symbolic calculations with Python.

In the lesson that precedes the current one, we look at how the inputs regarding the means and the covariance matrix of returns had a significant impact on the computation of the efficient frontier and the optimal portfolio itself. This lessons looks at how to extend the optimization program to make the problem of identifying the best portfolio more suitable for real life applications.

1. Generalities

Optimizing a portfolio with respect to mean and variance is rather simplistic. As shown in Figure 1, the number of inputs considered is very limited.

Figure 1. Mean-variance optimization and how it works.



There are several criticisms of this technique. It does have the merit of being convenient, but it does not address several issues:

- quality of the inputs (which we discussed in the previous lesson);
- concentrated asset class allocation (also discussed already in Module 1);
- skewness and kurtosis (considered in this lesson);
- risk-diversification (asset classes could be driven by same factors, i.e., share some of the same sources of risk);
- ignores liabilities (sometimes investors need cash to pay for certain liabilities they have in the future);
- single period timeframe (it does not take into account cash flows over time or the serial correlation of returns among different asset classes).

In this lesson, we will look specifically at how to deal with skewness and kurtosis, and we will also consider the issue of liquidity. This will bring us to a multi-criteria objective function, sort of, where more than just expected returns and volatilities are of concern. We will also add additional criteria to our optimization problem.

We must be clear about one aspect: It is impossible to exhaust the topic that has been the subject of entire master's projects and doctoral theses; nevertheless, it is important for practitioners to have a thorough understanding of these issues.

2. Coskewness

Computing the skewness of a portfolio directly from, say, the daily returns or the monthly returns of the portfolio itself is enough to assess certain characteristics of the portfolio. However, if we wanted to assess the impact of different single assets on the overall skewness of a portfolio, we would need to introduce the concept of **coskewness**. We will spend a fair amount of time on coskewness, but it will allow us to move much quicker when dealing with cokurtosis in the next section.

Simply put, coskewness is the ***third standardized cross central moment***. In more practical terms, it provides a measure of how much three random variables change together (we know already that the covariance is a measure of how two random variables change together, instead). The formal definition for random variables is the following:

$$\gamma_{XYZ} = \frac{E[(X-\mu_X)\cdot(Y-\mu_Y)\cdot(Z-\mu_Z)]}{\sigma_X \cdot \sigma_Y \cdot \sigma_Z}$$

where $\mu_X, \mu_Y,$ and μ_Z are the expected values of the random variables $X, Y,$ and $Z,$ respectively. Notice that if $X = Y = Z,$ the formula above becomes:

$$\gamma_{XXX} = \frac{E[(X-\mu_X)^3]}{\sigma_X^3},$$

i.e., the usual measure of skewness for the case of an individual random variable. We use the symbol γ because s or σ are already used for variances and covariances.

We will need to sweat it a bit and earn a deeper understanding of what coskewness is all about and how to represent it. For instance, we also need to deal with elements like

$$\gamma_{XXY} = \frac{E[(X-\mu_X)^2\cdot(Y-\mu_Y)]}{\sigma_X^2 \cdot \sigma_Y}$$

etc. In an attempt to simplify the notation as much as possible, we use $M_i, i = 1, 2, 3, 4$ to denote the i-th central moment matrix of returns. Thus, $M_2 = \Sigma,$ the variance-covariance matrix, which we know to contain moments like $E[X-\mu_X]^2, E[Y-\mu_Y]^2,$ and $E[X-\mu_X]E[Y-\mu_Y].$ Instead, M_3 is used to represent the coskewness matrix. Later, we will see that M_4 will be adopted as a symbol for the cokurtosis matrix.

The notation becomes quickly complicated. However, in the case of a 2-asset portfolio we have

$$M_3 = \begin{pmatrix} \gamma_{111} & \gamma_{112} & \gamma_{121} & \gamma_{122} \\ \gamma_{211} & \gamma_{212} & \gamma_{221} & \gamma_{222} \end{pmatrix}.$$

We should notice at least three things about this matrix:

- It is a $2 \times 2^2 = 2 \times 4$ matrix.
- It contains only four distinct element since $\gamma_{112} = \gamma_{121} = \gamma_{211}, \gamma_{122} = \gamma_{212} = \gamma_{221}.$
- In general, in the case of an m-asset portfolio, the matrix M is going to be a $m \times m^2$ matrix consisting of $m(m+1)(m+2)/6$ distinct elements.

Example. In the case of the $S\&P500$ index, M_3 would consist of $500 \cdot 500^2 = 125,000,000$ elements of which $20,968,500$ are distinct. So, computationally speaking, this is not a small task.

Why all this effort? Well, we need to be able to express the skewness of an m-asset portfolio as a function of the weights of each of the individual assets. If we are dealing with an m-asset portfolio and we describe it by the system of weights $w^T = (w_1, w_2, \ldots, w_m),$ the skewness of the overall portfolio has the following formula:

$$\text{skewness of portfolio } \equiv \gamma = w^T M_3 (w \otimes w).$$

The notation $w \otimes w$ is called a **tensor product** or a **Kronecker product** and it means the following:

$$w \otimes w = \begin{pmatrix} w_1 \cdot w_1 \\ w_1 \cdot w_2 \\ w_2 \cdot w_1 \\ w_2 \cdot w_2 \end{pmatrix}.$$

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There is nothing mysterious about this, and it is simply the consequence of how Kronecker products work. In general, if a is a $m \times n$ matrix and B is a $p \times q$ matrix, then

$$A \otimes B = \begin{pmatrix} a_{11}B & a_{12}B & \dots & a_{1n}B \\ a_{21}B & a_{22}B & \dots & a_{2n}B \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1}B & a_{m2}B & \dots & a_{mn}B \end{pmatrix}$$

which is an $mp \times nq$ matrix. in our case $A = w$ a $n \times 1$ matrix and $B = w$ a $n \times 1$ matrix so that $w \otimes w$ is a $2 \cdot 2 \times 1 \cdot 1 = 4 \times 1$ matrix.

To get more acquainted with the notion, we compute the skewness of a 2-asset portfolio using the formula provided:

$$\begin{aligned} \text{skewness}_p \equiv \gamma_p &= w^T M_3 (w \otimes w) \\ &= (w_1, w_2) \cdot \begin{pmatrix} \gamma_{111} & \gamma_{112} & \gamma_{211} & \gamma_{212} \\ \gamma_{121} & \gamma_{122} & \gamma_{221} & \gamma_{222} \end{pmatrix} \cdot \begin{pmatrix} w_1 \cdot w_1 \\ w_1 \cdot w_2 \\ w_2 \cdot w_1 \\ w_2 \cdot w_2 \end{pmatrix} \\ &= \gamma_{111} w_1^3 + (\gamma_{121} + \gamma_{112} + \gamma_{211}) w_1^2 w_2 + (\gamma_{122} + \gamma_{221} + \gamma_{212}) w_1 w_2^2 + \gamma_{222} w_2^3. \end{aligned}$$

We could simplify it even further by recalling that the terms inside each group of parentheses are all the same. We are starting to see how the skewness of a portfolio relates to the weights and the coskewness of individual assets.

Luckily, we do not need to do this by hand. Some students may feel intimidated by the use of tensor notation/Kronecker products, but most relevant software has Kronecker products built in. For instance, Python has the function **numpy.kron** while R has the product $\%x\%$, instead.

Example (Kronecker products in Python). Say that we wanted to perform the Kronecker product of two vectors $a^T = (1, 2, 3)$ and $b^T = (0, 1, 1)$. We can have Python do it for us as follows:

```
import numpy as np
a = np.array([[1],[2],[3]])
b = np.array([[0],[1],[1]])
result = np.kron(a, b)
print(" a kronecker b = ")
print(result)

a kronecker b =
[[0]
[1]
[1]
[0]
[2]
[2]
[0]
[3]
[3]]
```

For the Kronecker product of two or more matrices, things are not really different. But there would be potentially a large output. For instance,

```
A = [[1, 2], [0,2]]
B = [[1,1], [1,-1]]
C = np.kron(A, B)
print(C)
```

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```
[[ 1  1  2  2]
[ 1 -1  2 -2]
[ 0  0  2  2]
[ 0  0  2 -2]]
```

Example. (Computing skewness of a portfolio from coskewness of individual assets). A portfolio consists of two assets and the weights of each asset are $w^T = (0.4, 0.6)$. The co-skewness matrix is the following:

$$M_3 = \begin{pmatrix} 0.2 & 0.1 & 0.1 & 0.3 \\ 0.1 & 0.3 & 0.3 & 0.5 \end{pmatrix}.$$

Find the skewness of the portfolio.

The formula is the one given earlier, i.e.:

$$\text{skewness}_p = w^T M_3 (w \otimes w).$$

This could still be done by hand using the formula shown earlier. However, we can also use Python to do it for us.

```
import numpy as np
w = np.array([[0.4],[0.6]])
M =np.array([[0.2, 0.1, 0.1, 0.3],[0.1,0.3,0.3,0.5]])
Sk = np.matmul(np.matmul(np.transpose(w), M), np.kron(w,w))
print(Sk)
[[0.2792]]
```

i.e. the skewness of the portfolio is 0.2792.

Before we move forward, it is useful to think about a few relevant takeaways:

- **Positive coskewness** means that the three stocks will tend to undergo extreme deviations at the same time (but notice that they could be all three movements in the positive direction or 2 in the negative direction and 1 in the positive direction)
- **Negative coskewness** also means that three stocks will tend to undero extreeme movements at the same time and either all 3 moves in the negative direction or one moves in the negative direction and two in the positive direction.

For a portfolio it is difficult to quantify what the actual practical implications are. Of course we like when positive coskewness means that all three stocks will move in the positive direction, but we like it less when the positive coskewness is the consequence of two stocks undergoing large negative movements and only one exhibits a large positive movement. Thomas Kim in a paper quoted in the reference looked into this problem and found that, while high coskewness stocks tend to increase the skewness of the portfolio (a good thing), the number of assets had a larger impact on the skewness of the portfolio than the individual assets' coskewnesses and a large number of assets tends to eliminate the impact of high positive coskewness assets. But it is not a conclusive study.

Can we represent coskewness graphically? Definitely not as easily or effectively as the covariance. However, the next example, taken from Wikipedia, supplies all the calculations and a couple of plots to it (see *coskewness from Wikipedia*). The gist of this example can be summarized like this: *We can start with two random variables, which individually have no skewness. However, unless the two random variables are independent, their sum can have either negative or positive skewness (depending on the correlation).*

Students who are interested can find the details below. Some mathematical sophistication is required, but ambitious quants should definitely welcome the challenge.

Example (Skewness and coskewness. OPTIONAL). Let X be a standard normal random variable and let Y be another random variable defined as follows:

$$Y \sim \begin{cases} X & \text{if } X < 0; \\ T & \text{if } X \geq 0; \end{cases}$$

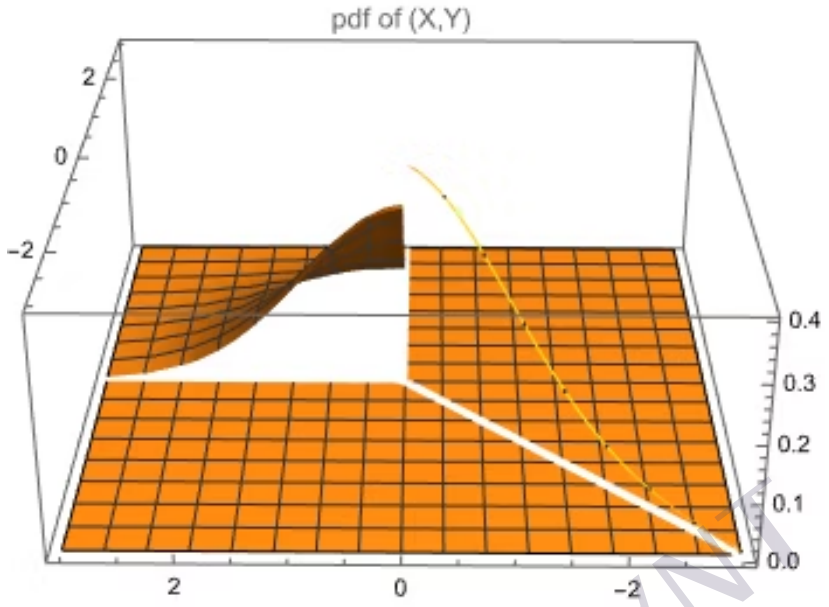
where T is a half-normal random variable defined on the positive half of the real line and independent of X.

Then, the joint pdf of X and Y is given by

$$f_{X,Y}(x,y) = \frac{e^{-x^2/2}}{\sqrt{2\pi}}\left(\delta(x-y)\cdot 1_{x<0} + 2\frac{e^{-y^2/2}}{\sqrt{2\pi}}\cdot 1_{x\geq 0}\cdot 1_{y\geq 0}\right)$$

where $\delta(x)$ is the Dirac delta function. The picture of the joint density is shown in Figure 2 below. Note that the visual effect along the diagonal in the negative quadrant has been magnified so to be evident in the picture.

Figure 2: The Joint pdf of X and Y.



Now, both X and Y are marginally $\mathcal{N}_{0,1}$ random variables. For instance:

$$f_Y(y) = \int_{-\infty}^{0} \frac{e^{-x^2/2}}{\sqrt{2\pi}}\delta(x-y)\,dx + \int_0^{\infty} \frac{e^{-x^2/2}}{\sqrt{2\pi}} 2\frac{e^{-y^2/2}}{\sqrt{2\pi}}\,dx.$$

For the first integral, we need to use the property of the Dirac delta function, i.e.,

$$\int_{\infty}^{\infty} f(y)\delta(x-y)\,dy = f(x).$$

In our case, this becomes:

$$\int_{-\infty}^{0} \frac{e^{-x^2/2}}{\sqrt{2\pi}}\delta(x-y)\,dx = \frac{e^{-y^2/2}}{\sqrt{2\pi}}\cdot 1_{\{y<0\}}.$$

The other integral is easier to handle and yields

$$\int_0^{\infty} \frac{e^{-x^2/2}}{\sqrt{2\pi}} 2\frac{e^{-y^2/2}}{\sqrt{2\pi}}\,dx = \frac{e^{-y^2/2}}{\sqrt{2\pi}}\cdot 1_{\{y\geq 0\}}$$

and which proves that Y is marginally a $\mathcal{N}_{0,1}$ random variable as claimed.

As both X and Y are $\mathcal{N}_{0,1}$ random variables, neither has any skewness.

However, it is clear that X and Y are not independent (why?), and their sum turns out to be negatively skewed. This can be shown in two ways (which are great exercises in master's level statistics).

The first method uses a change of variables: Let $U = X + Y$ and $V = X$. Then $X = V, Y = U - V$, and

$$\begin{aligned} f_U(u) &= \int_{-\infty}^{0} \frac{e^{-v^2/2}}{\sqrt{2\pi}}\delta(2v-u)\cdot 1_{\{u<0\}}\,dv + 2\int_0^{\infty} \frac{e^{-v^2/2}e^{-(u-v)^2/2}}{2\pi}\cdot 1_{\{u\geq v\}}\,dv \\ &= \int_{-\infty}^{0} \frac{e^{-v^2/2}}{\sqrt{2\pi}}\frac{1}{2}\delta\left(v-\tfrac{u}{2}\right)\cdot 1_{\{u<0\}}\,dv + \frac{e^{-u^2/2}}{\pi}\int_0^{\infty} e^{-v^2/2+uv}\cdot 1_{\{u\geq v\}}\,dv. \end{aligned}$$

Now, from the properties of the delta function, we have that the first integral is equal to

$$\frac{e^{-(u/2)^2/2}}{2\sqrt{2\pi}}\cdot 1_{u<0}.$$

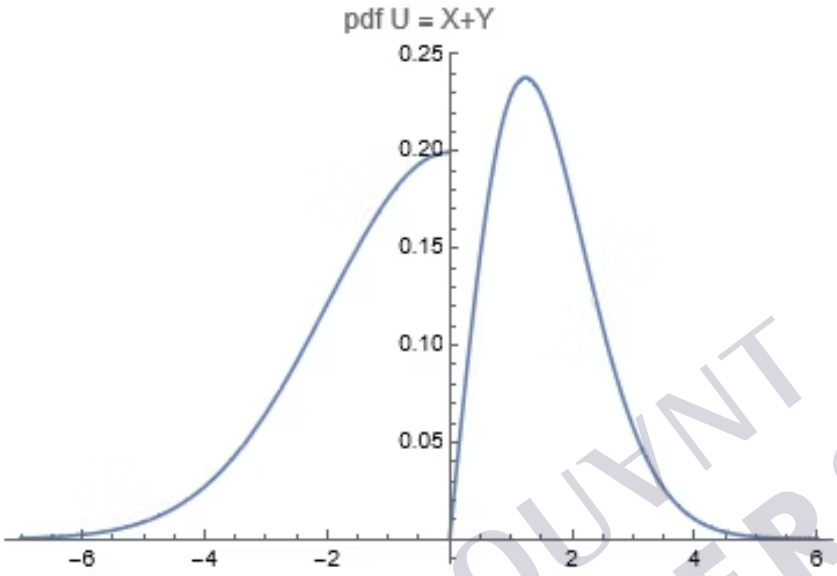
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The second integral, if we add and subtract $u^2/4$ to the power of the exponential function, after a few algebraic manipulations, should be

$$\frac{e^{-u^2/4}}{2\sqrt{\pi}}\int_0^u \frac{2}{\sqrt{\pi}}e^{-(v-u/2)^2}\,dv = \frac{e^{-u^2/4}}{2\sqrt{\pi}}\int_{-u/2}^{u/2}\frac{2}{\sqrt{\pi}}e^{-(w)^2}\,dw$$
$$= \frac{e^{-u^2/4}}{2\sqrt{\pi}}\cdot 2\text{erf}(u/2)\cdot 1_{\{u\geq 0\}} = \frac{e^{-u^2/4}}{\sqrt{\pi}}\cdot \text{erf}(u/2)\cdot 1_{\{u\geq 0\}}.$$

The pdf of the sum of X and Y is shown in Figure 3 and it displays some negative skewness, despite neither X nor Y having any skewness of their own. Dependence is what is causing it.

Figure 3: The Distribution of $X+Y$.



A second way to establish this result is to use the definition of coskewness directly. Simple algebra shows that

$$\gamma_{X+Y} = \frac{1}{\sigma^3_{X+Y}}\cdot\left(\sigma^3_X\gamma_{XXX} + 3\sigma^2_X\sigma_Y\gamma_{XXY} + 3\sigma_X\sigma^2_Y\gamma_{XYY} - \sigma^3_Y\gamma_{YYY}\right).$$

We know already that $\gamma_{XXX} = \gamma_{YYY} = 0$ because the two marginal distributions are standard normal. We also know that $\sigma^2_X = \sigma^2_Y = 1$ (both marginal distributions are $N(0,1)$). Furthermore, since $\mu_X = 0$ and $\mu_Y = 0$,

$$\gamma_{XXY} = \int_{-\infty}^0\int_{-\infty}^0 x^2 y\frac{e^{-x^2/2}}{\sqrt{2\pi}}\delta(x-y)\,dxdy + \int_0^\infty\int_0^\infty 2\frac{e^{-x^2/2}}{\sqrt{2\pi}}\frac{e^{-y^2/2}}{\sqrt{2\pi}}\,dxdy$$
$$= \int_{-\infty}^0 x^2\frac{e^{-x^2/2}}{\sqrt{2\pi}}\,dx\int_{-\infty}^0 y\delta(x-y)\,dy + \int_0^\infty 2x^2\frac{e^{-x^2/2}}{\sqrt{2\pi}}\,dx\int_0^\infty\frac{e^{-y^2/2}}{\sqrt{2\pi}},dy$$
$$= \int_{-\infty}^0 x^3\frac{e^{-x^2/2}}{\sqrt{2\pi}}\,dx + \int_0^\infty 2x^2\frac{e^{-x^2/2}}{\sqrt{2\pi}}\,dx\int_0^\infty\frac{e^{-y^2/2}}{\sqrt{2\pi}}\,dy,$$

using once more the property of the Dirac delta function for the first integral. At this point, the first integral yields

$$\int_{-\infty}^0 x^3\tfrac{e^{-x^2/2}}{\sqrt{2\pi}}\,dx = -\int_{-\infty}^0\tfrac{x^2}{\sqrt{2\pi}}\tfrac{de^{-x^2/2}}{dx}\,dx$$

which can be integrated by parts and yields

$$\int_{-\infty}^0 x^3\frac{e^{-x^2/2}}{\sqrt{2\pi}}\,dx = -\frac{\sqrt{2}}{\sqrt{\pi}}.$$

The second integral is separable, and, using the moments of the half-normal distribution, we find that

$$\int_0^\infty 2x^2\frac{e^{-x^2/2}}{\sqrt{2\pi}}\,dx\int_0^\infty\frac{e^{-y^2/2}}{\sqrt{2\pi}}\,dy = \frac{1}{\sqrt{2\pi}}.$$

Thus, putting everything together, we have shown that $\gamma_{XXY} = -\frac{1}{\sqrt{2\pi}}$. Similarly, one shows that $\gamma_{XYY} = -\frac{1}{\sqrt{2\pi}}$, as well. Furthermore, we can show that $Cov(X,Y) = \frac{\pi+2}{2\pi}$ and from that, $\sigma_{X+Y} = \sqrt{\frac{3\pi+2}{\pi}}$. Finally:

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$$\gamma_{X+Y} = -\frac{3\sqrt{2}\pi}{(2+3\pi)^{3/2}} \approx -0.345.$$

In other words, we have produced an example where, starting with two normal random variables with zero skewness, their sum exhibits instead negative skewness due to the nature of the correlation between the two original random variables. That is why coskewness is relevant!

Other more useful discussions and some visuals about the types of coskewness and cokurtosis can be found in the following reading: Least Quartic Regression Criterion to Evaluate Systematic Risk in the Presence of Co-Skewness and Co-Kurtosis

We have learned a lot about coskweness so far and how it relates to skewness of a portfolio starting from the individual assets. However, we also saw that the coskweness matrix needed, M_3, can be really a big matrix to compute even if many elements are the same. Thus, M_3 has to be estimated from the returns data of the individual assets over the reference period.

How do we go from the theoretical formula for coskewness for random variables to one that uses the sample data? For instance:

$$\widehat{\gamma}_{112} = \frac{\sum_{t=1}^{N}(r_{1t} - \bar{r}_1)(r_{1t} - \bar{r}_1)(r_{2t} - \bar{r}_2)}{(N-1)s_1 s_1 s_2} = \frac{\sum_{t=1}^{N}(r_{1t} - \bar{r}_1)^2(r_{2t} - \bar{r}_2)}{(N-1)s_1^2 s_2},$$

etc.

3. Cokurtosis

The previous section was quite long because we discussed the issues related to the concept of coskewness. The same basic ideas apply to the notion of cokurtosis. **Cokurtosis** deals with the standardized 4-th central moments of random variables. Unlike covariance, which involves two random variables at a time, or coskewness, which involves three random variables at a time, cokurtosis uses four variables at the same time.

We will use the symbol $M4$ for the kurtosis matrix.

The concept of cokurtosis is important for applications in portfolio theory because if two random variables or assets exhibit a high level of cokurtosis, then they will move together undergoing either large positive or large negative deviations during the assigned reference period. This obviously has implications for when one uses VaR to assess the risk of portfolios since coskweness and cokurtosis would tend to produce distributions with heavy tails and/or that are skewed.

The main practical complication stems from the fact that when using fourth moments, there are many of them.

In the case of four random variables, $W, X, Y,$ and $Z,$ then cokurtosis is defined by

$$K_{WXYZ} = \frac{E[(W - \mu_w) \cdot (X - \mu_X) \cdot (Y - \mu_Y) \cdot (Z - \mu_Z)]}{\sigma_W \sigma_X \sigma_Y \sigma_Z}$$

and all its possible variations like for example

$$K_{XXYZ} = \frac{E[(X - \mu_X)^2 \cdot (Y - \mu_Y) \cdot (Z - \mu_Z)]}{\sigma_X^2 \sigma_Y \sigma_Z},$$

etc. In particular when $W = Y = Z = X$ we have the traditional definition of kurtosis for a random variable, that is:

$$K_X = \frac{E[(X - \mu_X)^4]}{\sigma_X^4}.$$

For applications, it is more important to use the sample formula instead of the theoretical one for random variables/populations as we will be using samples of returns from assets. To this purpose, if we consider the returns for four assets in the portfolio, say $r_h, r_i, r_j, r_k,$ over a period of, say N- days, then the sample cokurtosis for the returns of these 4 assets is given by:

$$\widehat{K}_{hijk} = \frac{N(N+1)(N-1)}{(N-2)(N-3)} \cdot \frac{\sum_{t=1}^{N}(r_{ht} - \bar{r}_h)(r_{it} - \bar{r}_i)(r_{jt} - \bar{r}_j)(r_{kt} - \bar{r}_k)}{(\sum_{t=1}^{N}(r_{ht} - \bar{r}_h)^2)^{1/2} \cdot (\sum_{t=1}^{N}(r_{it} - \bar{r}_i)^2)^{1/2} \cdot (\sum_{t=1}^{N}(r_{jt} - \bar{r}_j)^2)^{1/2} \cdot (\sum_{t=1}^{N}(r_{kt} - \bar{r}_k)^2)^{1/2}}$$
$$\approx \frac{N \cdot \sum_{t=1}^{N}(r_{ht} - \bar{r}_h)(r_{it} - \bar{r}_i)(r_{jt} - \bar{r}_j)(r_{kt} - \bar{r}_k)}{(\sum_{t=1}^{N}(r_{ht} - \bar{r}_h)^2)^{1/2} \cdot (\sum_{t=1}^{N}(r_{it} - \bar{r}_i)^2)^{1/2} \cdot (\sum_{t=1}^{N}(r_{jt} - \bar{r}_j)^2)^{1/2} \cdot (\sum_{t=1}^{N}(r_{kt} - \bar{r}_k)^2)^{1/2}}$$

if N is large. The last formula can be rewritten in a more compact manner as:

$$\frac{1}{N-1}\frac{\sum_{t=1}^{N}(r_{ht}-\bar{r}_h)(r_{it}-\bar{r}_i)(r_{jt}-\bar{r}_j)(r_{kt}-\bar{r}_k)}{s_h s_i s_j s_k} \quad \text{if N is large}$$

The cokurtosis matrix,M_4, is now very large. For just two assets, it looks like this:

$$M_4 = \begin{pmatrix} K_{1111} & K_{1112} & K_{1121} & K_{1122} & K_{1211} & K_{1212} & K_{1221} & K_{1222} \\ K_{2111} & K_{2112} & K_{2121} & K_{2122} & K_{2211} & K_{2212} & K_{2221} & K_{2222} \end{pmatrix}.$$

Most of these elements are the same, but we should notice that even for a two-asset problem,M_4 is a$2 \times 2^3 = 8$ matrix. In case we were working with m assets, the number of elements in M_4 would become a $m \times m^3$ matrix, which can be quite large in the case of index portfolios. For the S&P 500, it would be a matrix of dimensions $500 \times 500^3 = 500 \times 125,000,000$! However, one could check using combinatorics (try counting the number of unordered arrangements of size r from m object while sampling with replacement), that the number of distinct entries is only:

$$1 + \binom{m-1}{1} + \binom{m}{2} + \binom{m+1}{3} + \binom{m+2}{4}.$$

The kurtosis of a portfolio with weights $w^T = (w_1, w_2, \ldots, w_m$ is obtained as:

$$\text{kurtosis of portfolio } = w^T M_4 (w \otimes w \otimes w).$$

4. Optimal Portfolios Adding Skewness and Kurtosis

Let's start by looking at a couple of plots involving our portfolio of eight assets that we have used many times already in previous lessons.

Figure 4 shows 10,000 simulated portfolios from eight assets. The portfolios close to the efficient frontier do not have all the same skewness, and negative skewness between -0.1 and -0.2 appears the most characteristic value.

Figure 4: 10,000 Simulated Portfolios and Their Skewness

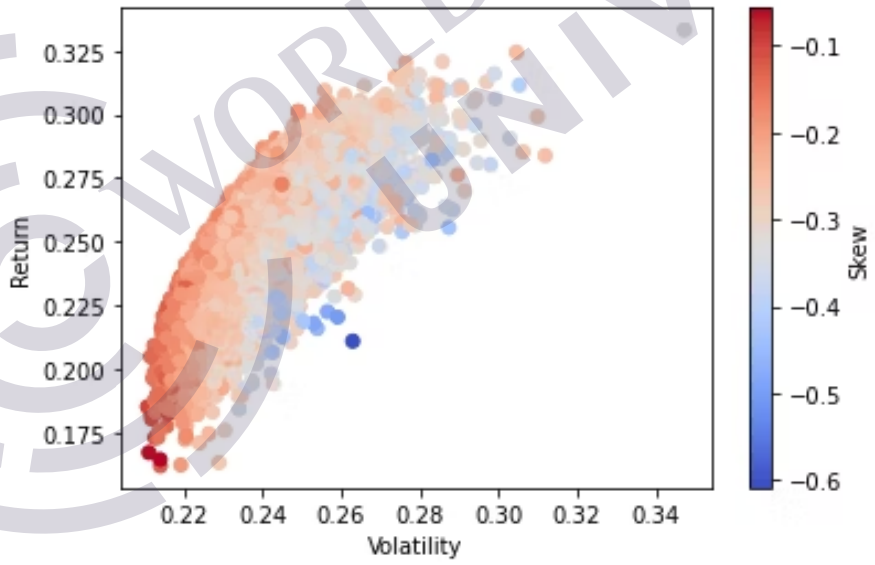
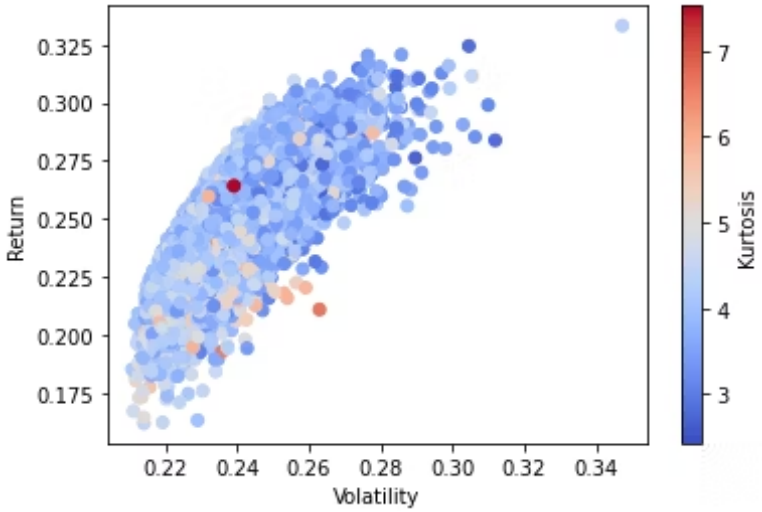


Figure 5 below also shows the results of the same 10,000 simulated portfolios, but in this case, we look at the kurtosis of those portfolios. The majority of these portfolios have kurtosis between 4 and 5, i.e., heavier tails than in the case of a normal distribution.

Figure 5: 10,000 Simulated Portfolios and Their Kurtosis

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Considering the aspects highlighted in the two figures immediately above, it makes sense that to protect portfolios from risk, while trying to optimize the risk-return trade-off, skewness and kurtosis should be introduced in the optimization program. Hence, our optimization program becomes:

$$\max_{w} \alpha_1 w^T M_1 + \alpha_2 w^T M_2 w + \alpha_3 w^T M_3 (w \otimes w) + \alpha_4 w^T M_4 (w \otimes w \otimes w)$$
$$\text{subject to} \quad w^T \iota = 1.$$

Note that we wrote the program using the matrices/arrays of moments to make it clear that we are now using additional moments compared with what we did in earlier lessons.

$\alpha_1, \alpha_2, \alpha_3,$ and α_4 are parameters that control how important that particular moment is. While the magnitude of those parameters can potentially change among individuals, it seems quite realistic to assume that $\alpha_1 > 0$,$\alpha_2 < 0$ (economic agents would prefer to avoid volatility/risk as much as possible, but with differing degrees among different individuals),$\alpha_3 > 0,$ and $\alpha_4 < 0.$ The rationales are that:

- $\alpha_1 > 0,$ since economic agents want a high expected return;
- $\alpha_2 < 0,$ as agents would prefer to avoid volatility/risk as much as possible, but with differing degrees among different individuals;
- $\alpha_3 > 0,$ because economic agents welcome positive skewness but dislike negative skewness;
- $\alpha_4 < 0,$ because higher kurtosis has the ability to cause large losses (as well as large gains). The difference among individuals in fearing kurtosis is modeled via the size of the parameter as is the case for the other three as well.

Needless to say, we can add many more constraints to our optimization program. Now, it should also be completely clear why we devoted a considerable amount of time to the notions of coskewness and cokurtosis: We needed to express those statistics as a function of the weights of the individual assets that are/could be considered in forming our portfolio. Now, we have an optimization problem.

5. Liquidity Constraints

Most would agree that liquidity can be defined as the efficiency or ease with which an asset or security can be converted into ready cash without affecting its market price. It is a very desirable feature, and in addition, liquid assets imply an awareness that one can buy or sell assets within fair price levels. Liquidity can be characterized in more than one way.

Securities that are fungible, i.e. mutually interchangeable, are more likely to be liquid than securities, for which no two are exactly alike. For instance, homes are not necessarily fungible in general (due to different locations, sizes, etc.). Expensive arts are likely illiquid assets as one has to find suitable buyers with both the interest and the money to purchase the assets. At the opposite extreme, the stock market is characterized by higher market liquidity. In particular, whenever an exchange has a high volume of trade that is not dominated by selling, the price a buyer offers per share (bid price) and the price the seller is willing to accept (ask price) will be fairly close to each other.

There is more than one way to measure the liquidity of an asset/investment: bid-ask spread, as we just mentioned in the previous paragraph; volume-related liquidity measures such as the volume or quantity of shares traded or the number of transactions/orders per unit of time; or turnover ratio, that is the proportion of shares of a stock traded during a specific time reference (say one month) out of the outstanding number of shares of that particular security. Often, the type of assets dictate what measure one can use. We will adopt a turnover ratio as a measure of liquidity, but the student should be aware that we are

not committed to a particular measure of liquidity; we are just concentrating on how to use liquidity in our portfolio selection process.

Once we have calculated the liquidity measure of each asset/stock, say $L_i, i = 1, 2, \ldots, m,$ the higher the number L_i is, the more liquid an asset i would be considered. For a portfolio, we could define liquidity as

$$L_{p.folio} = \sum_{i=1}^{m} w_i L_i \quad \text{or} \quad L_{p.folio} = w^T L,$$

where $L^T = (L_1, \ldots, L_m).$

This seems like a plausible measure of liquidity for a portfolio, and we can conjecture that the first moment of liquidity (mean) is important in portfolio optimization. However, there is nothing preventing us from using higher moments for liquidity as we did for returns. In fact, the individual liquidity of certain assets may **co-move**, i.e., in certain periods, all assets can trade very liquid and in others, many assets' volumes could dry up. It is also possible that when certain assets are very liquid, others see very limited trading. Since liquidity can drive returns, liquidity could enter as an element of our objective function. In addition, if the co-movement of liquidity does not follow patterns similar to those of returns, it makes sense to consider higher moments of liquidity in our objective function as well.

This leads us to expand our optimization problem to look like this:

$$\begin{aligned} \max_{w} \alpha_1 w^T M_1 + \alpha_2 w^T M_2 w + \alpha_3 w^T M_3 (w \otimes w) + \alpha_4 w^T M_4 (w \otimes w \otimes w) + \\ \beta_1 w^T M_1^L + \beta_2 w^T M_2^L w + \beta_3 w^T M_3^L (w \otimes w) + \beta_4 w^T M_4^L (w \otimes w \otimes w) \\ \text{subject to} \quad w^T \iota = 1, \end{aligned}$$

with $M_i^L, i = 1, 2, 3, 4$ representing the central moments for liquidity instead of returns and $\beta_i, i = 1, 2, 3, 4$ representing a specific investor's preferences. For instance, some investors could prefer higher moments while others may be uneasy and penalize the higher moments.

Also, we could rewrite the problem above as

$$\begin{aligned} \max_w \lambda_R \big(\alpha_1 w^T M_1 + \alpha_2 w^T M_2 w + \alpha_3 w^T M_3 (w \otimes w) + \alpha_4 w^T M_4 (w \otimes w \otimes w)\big) + \\ \lambda_L \big(\beta_1 w^T M_1^L + \beta_2 w^T M_2^L w + \beta_3 w^T M_3^L (w \otimes w) + \beta_4 w^T M_4^L (w \otimes w \otimes w)\big) \\ \text{subject to} \quad w^T \iota = 1, \end{aligned}$$

where the parameters λ_R and λ_L are modeling specific preferences for returns and/or liquidity.

This has led us to consider multiple criteria in selecting an optimal portfolio, and we can definitely add many more constraints. Hence, we believe that students who followed the exposition so far are now in a good place to appreciate several subtle nuances that come with portfolio optimization theory.

6. General Optimization Techniques

A problem like

$$\begin{aligned} \max_{w} \alpha_1 w^T M_1 + \alpha_2 w^T M_2 w + \alpha_3 w^T M_3 (w \otimes w) + \alpha_4 w^T M_4 (w \otimes w \otimes w) \\ \text{subject to} \quad w^T \iota = 1. \end{aligned}$$

is not an instance of quadratic programming (QP) anymore. We are still dealing with a polynomial objective function, but the variables $w_1, \ldots, w_m$ appear with powers higher than two. This means that one has to use more powerful techniques that can handle this type of problem, and we have to be aware that nonlinear constrained optimization is not an easy task in general. In fact, it could be a topic for an entire class on its own. Python provides several packages to solve these problems. We have neither the time nor space to investigate this topic in full, but we will look at how to deal with problems like the one at hand.

Specifically we will use the **minimize** package from the **scipy.optmize** library with the *SLSQP optimizer* (an optimizer based on a sequential least squares programming algorithm), and we will show how to use it by solving a specific problem that incorporates all relevant features.

$$\begin{aligned} &\min_{w_1, w_2, w_3, w_4} w_1 \cdot w_4 \cdot (w_1 + w_2 + w_3) + w_4 \ \ \text{(objective function)} \\ &\text{s.t.} \ \ w_1 w_2 w_3 w_4 \geq 25 \ \ \text{(inequality constraint)} \\ &\qquad w_1^2 + w_2^2 + w_3^2 + w_4^2 = 45 \ \ \text{(equality constraint)} \\ &\qquad 1 \leq w_1, w_2, w_3, w_4 \leq 5 \ \ \text{(bounds for the variables)} \\ &\qquad w_0 = (1, 6, 3, 2) \ \ \text{(initial guess for the solutions)} \end{aligned}$$

The Python code to solve this problem is the following:

```
import numpy as np
from scipy.optimize import minimize  # load the package minimize

def objective(w):  # define the objective function to minimize.
   return w[0]*w[3]*(w[0]+w[1]+w[2])+w[3]

def constraint1(w):   # define the typical inequality constraint
   return w[0]*w[1]*w[2]*w[3]-25.0

def constraint2(w):   # define the prototypical equality constraint.
   return w[0]**2 + w[1]**2 + w[2]**2 + w[3]**2 - 45.0

# initial weights
w0 = (1,6,3,2)

# show initial objective using the initial solution
print('Initial Objective: '+ str(objective(w0)))

# optimize
b = (1.0,5.0)
bnds = (b, b, b, b)   # define the interval bounds for each of the variables.
con1 = {'type': 'ineq', 'fun': constraint1}  # tell the program the type of constr.
con2 = {'type': 'eq', 'fun': constraint2}   # tell the program the type of constr.
cons = ([con1,con2])    # this is the list of constraints the program will receive
solution = minimize(objective,w0,method='SLSQP', bounds=bnds,constraints=cons)
w = solution.x

# show final objective
print('Final Objective: ' + str(objective(w)))

# print solution
print('Solution')
print('w1 = ' + str(w[0]))
print('w2 = ' + str(w[1]))
print('w3 = ' + str(w[2]))
print('w4 = ' + str(w[3]))
```

If we want to maximize a function, say $f(w_1, w_2, w_3)$, we will do so by minimizing $-f(w_1, w_2, w_3)$. Note the particular form of the inequality constraint. If we wanted $\leq$ instead of $\geq$, again, we can do that just by multiplying both sides of the inequality by -1.

Finally, one caveat: Non-linear constrained optimization is complex and the algorithm may not find the optimal global solution and could instead end up on a local solution. Sometimes, changing the initial condition could tell us whether that is the case or not.

Now we go back to our portfolio optimization problem, and this time, we add skewness and kurtosis to the mean and variance. The first task is to be able to write the objective function. To keep things reasonably simple, we are going to consider only three stocks and not eight as before. The objective function consists of four parts:

```
import sympy as sy
import numpy as np
import pandas as pd
import math
import yfinance as yf
from datetime import datetime
import matplotlib.pyplot as plt
import seaborn as sns
from pandas_datareader import data

### define the stocks from which we want the data (via tickers) ###
#assets = ["AAPL", "MSFT","AMZN","GOOGL","META","TSLA","AMGN", "V"]
assets_list = ["AAPL", "MSFT", "AMZN"] # we worked with a reduced dataset to simplify
# Define the start date
start_date = "2015-01-2"
end_date = "2022-01-15"
# Today's date
#today = datetime.today().strftime('%Y-%m-%d')
# Create an empty dataframe
df_prices = pd.DataFrame()
# Get the  "Adj Close" data from Yahoo Finance
def YahooData(dataframe,assets_list,start_date,end_date):
    for i in assets_list:
        dataframe[i] = yf.download(i, start=start_date, end=end_date)['Adj Close']
    return dataframe
df1 = YahooData(df_prices,assets, start_date,end_date)
# We transform the data to it´s logarithmic returns
df = np.log(df1).diff()
# Drop the first row because we loose information we the logarithmic return.
df = df.dropna()

# compute coskewness matrix

num = len(df.columns)
numc = len(df)
M3 = np.zeros(shape = (num, num**2))
v = df.values
means = v.mean(0,keepdims=True)
v1 = (v-means).T

for i in range(num):
    for j in range(num):
        for k in range(num):
            u = 0.0
            for t in  range(numc):
                u=u+v1[i,t]*v1[j,t]*v1[k,t]

            M3[i, j*num+k] = u/float((len(df)-1)\
            *df.iloc[:,i].std()*df.iloc[:,j].std()*df.iloc[:,k].std())
```

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```
M3

# Compute the cokurtosis matrix

num = len(df.columns)
numc = len(df)
M4 = np.zeros(shape = (num, num**3))
v = df.values
means = v.mean(0,keepdims=True)
v1 = (v-means).T
for i in range(num):
    for j in range(num):
        for h in range(num):
            for k in range(num):
                u = 0.0
                for t in  range(numc):
                    u=u+v1[i,t]*v1[j,t]*v1[h,t]*v1[k,t]

                M4[i, j*num**2+ h*num+k] = u/float((len(df)-1)\
                *df.iloc[:,i].std()*df.iloc[:,j].std()*df.iloc[:,h].std()\
                *df.iloc[:,k].std())

M4

w1 = sy.Symbol('w1')
w2 = sy.Symbol('w2')
w3 = sy.Symbol('w3')
w =[w1,w2,w3]

M1 = np.round(v.mean(0,keepdims=True),6)
# define mean of portfolio
meanP = sy.simplify(np.dot(w, M1.T))

M2 = df.cov()

# define variance of portfolio.
varP = np.dot(np.dot(w, M2), w)
varP= sy.simplify(varP)

h = np.dot(w, M3)
k =np.kron(w,w)
# define skewness of portfolio
skewP =sy.simplify(np.dot(h,k))

p= np.kron(w, w)
q =np.kron(w, p)
r = np.dot(w, M4)
```

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```
# define kurtosis of portfolio
kurtP = sy.simplify(np.dot(r, q))
```

Then in our example this code produced the following results (students who run this code will find similar formulae with different numbers)

$$meanP = 0.001072 \times w_1 + 0.001099 \times w_2 + 0.001288 \times w_3$$

$$varP = 0.000335w_1^2 + 0.000428w_1w_2 + 0.000392w_1w_3 + 0.00029w_2^2 + 0.000422w_2w_3 + 0.000369w_3^2$$

$$\begin{aligned} skewP = {} & 0.27365w_1^3 - 1.11516w_1^2w_2 - 0.56307w_1^2w_3 - 0.9627w_1w_2^2 - 1.04442w_1w_2w_3 \\ & - 0.46659w_1w_3^2 - 0.2243w_2^3 - 0.02658w_2^2w_3 + 0.16305w_2w_3^2 + 0.52807w_3^3 \end{aligned}$$

and

$$\begin{aligned} kurtP = {} & 9.36158w_1^4 + 30.61548w_1^3w_2 + 17.86736w_1^3w_3 + 50.36082w_1^2w_2^2 + 51.23004w_1^2w_2w_3 \\ & + 19.875w_1^2w_3^2 + 38.88744w_1w_2^3 + 57.08256w_1w_2^2w_3 + 40.94832w_1w_2w_3^2 \\ & + 11.10984w_1w_3^3 + 13.61024w_2^4 + 26.82224w_2^3w_3 + 31.76712w_2^2w_3^2 + 17.96152w_2w_3^3 + 9.70931w_3^4 \end{aligned}$$

Now, we can set up the problem like in the case of the optimization one. The objective function will be quite complex, but not particularly difficult.

```
alpha1 = -1   # note! The signs have been changed to switch the
alpha2 = +1/2 # maximization program to a minimization one.
alpha3 = -1
alpha4 = 1
def mean(w):
return alpha1*(0.001072*w[0] + 0.001099*w[1] + 0.001288*w[2])

def var(w):
return alpha2*(0.000335*w[0]**2 + 0.000428*w[0]*w[1] + 0.000392*w[0]*w[2]+ 0.00029*w[

def skew(w):
return alpha3*(0.27365*w[0]**3 - 1.11516*w[0]**2*w[1] - 0.56307*w[0]**2*w[2]
- 0.9627*w[0]*w[1]**2 - 1.04442*w[0]*w[1]*w[2] - 0.46659*w[0]*w[2]**2
 - 0.2243*w[1]**3 - 0.02658*w[1]**2*w[2] + 0.16305*w[1]*w[2]**2 + 0.52807*w[2]**3)

def kurt(w):
 return alpha4*(9.36158*w[0]**4 + 30.61548*w[0]**3*w[1] + 17.86736*w[0]**3*w[2]
 + 50.36082*w[0]**2*w[1]**2 + 51.23004*w[0]**2*w[1]*w[2] + 19.875*w[0]**2*w[2]**2
 + 38.88744*w[0]*w[1]**3 + 57.08256*w[0]*w[1]**2*w[2] + 40.94832*w[0]*w[1]*w[2]**2
     + 11.10984*w[0]*w[2]**3 + 13.61024*w[1]**4 + 26.82224*w[1]**3*w[2]
     + 31.76712*w[1]**2*w[2]**2 + 17.96152*w[1]*w[2]**3 + 9.70931*w[2]**4)

def objective(w):
   return mean(w)+var(w)+skew(w) + kurt(w)

For the constraint, in this case

def constraint1(w):
   return w[0]+w[1]+w[2]-1.0

for the initial guess, instead, let's use the portfolio that assigns the same weight
```

```
n = 3
w0 = np.zeros(n)
w0[0] = 0.333
w0[1] = 0.333
w0[2] = 0.333

And now we can complete the code.

# show initial objective
print('Initial Objective: ' + str(objective(w0)))

b = (0,10000)  #short selling is not allowed
bnds = (b, b, b)
con1 = {'type': 'eq', 'fun': constraint1}
cons = ([con1])
solution = minimize(objective,w0,method='SLSQP',
                    bounds=bnds,constraints=cons)
w = solution.x

# show final objective
print('Final Objective: ' + str(objective(w)))

# print solution
print('Solution')
print('w1 = ' + str(w[0]))
print('w2 = ' + str(w[1]))
print('w3 = ' + str(w[2]))
```

Finally, the solutions can be calculated and we find:

```
Initial Objective: 5.379044158345797
Final Objective: 4.229455361815727
```

Solution

```
w1 = 0.45074095500925276
w2 = 1.1058862159352145e-17
w3 = 0.5492590449907473
```

7. True Multicriteria Setups

In what we did, skewness and kurtosis were added to the optimization program and, by changing the coefficients α_is in the objective function we can decide whether mean, variance, skewness, etc, should be equally relevant. That approach works, but we saw that the calculations can become very cumbersome and basically untenable for portfolios with many assets.

In a 2010 paper, Mhiri and Prigent have proposed a different and more application oriented approach to portfolio optimization with higher moments. More specifically, we refer the section of their paper that concernns Portfolio optimization based on Polynomial Goal Programming(Section 2.2) and here we describe their method. If we let $\overline{R}, V, S, K$ denote the expected return vector, the co-variance matrix, the co-skewness matrix and the co-kurtosis matric, respectively, then we can run four individual optimization problems in which each optimization ignores the other goals, (i.e. we treat each problem independently of the other three):

$$\begin{aligned}&\max R^{\star}(w) = w^T\overline{R}\\&\text{subject to: } w^T\iota = 1\\&\qquad\qquad w_1 \geq 0 \ldots, w_m \geq 0;\end{aligned}$$

$$\begin{aligned}&\min V^{\star}(w) = w^T V w\\&\text{subject to: } w^T\iota = 1\\&\qquad\qquad w_1 \geq 0 \ldots, w_m \geq 0;\end{aligned}$$

$$\begin{aligned}&\max S^{\star}(w) = w^T S(w \otimes w)\\&\text{subject to: } w^T\iota = 1\\&\qquad\qquad w_1 \geq 0, \ldots, w_m \geq 0;\end{aligned}$$

and

$$\begin{aligned}&\min K^{\star}(w) = w^T K(w \otimes w \otimes w)\\&\text{subject to: } w^T\iota = 1\\&\qquad\qquad w_1 \geq 0, \ldots, w_m \geq 0.\end{aligned}$$

Let $R^{\star}, V^{\star}, S^{\star},$ and $K^{\star}$ be the optimal solutions for each of the 4 problems above. We label these values aspired levels, but clearly we cannot achieve them all at
the same time, not in general at least. Then, we call $d_1, d_2, d_3,$ and d_4 the goal variables which account for the deviations of expected return, variance, skewness and kurtosis from the respective aspired levels. Now, we run the following problem:

$$\begin{aligned}&\min Z = \left|\frac{d_1}{R^{\star}}\right|^{\lambda_1} + \left|\frac{d_2}{V^{\star}}\right|^{\lambda_2} + \left|\frac{d_3}{S^{\star}}\right|^{\lambda_3} + \left|\frac{d_4}{K^{\star}}\right|^{\lambda_4}\\&\text{subject to: } w^t\overline{R} + d_1 = R^{\star}\\&\qquad\qquad w^T V w - d_2 = V^{\star}\\&\qquad\qquad w^T S(w \otimes w) + d_3 = S^{\star}\\&\qquad\qquad w^T K(w \otimes w \otimes w) - d_4 = K^{\star}\\&\qquad\qquad w^T\iota = 1\\&\qquad\qquad w_1 \geq 0, \ldots, w_m \geq 0\\&\qquad\qquad d_1 \geq 0, d_2 \geq 0, d_3 \geq 0, d_4 \geq 0.\end{aligned}$$

Here $\lambda_1, \lambda_2, \lambda_3,$ and λ_4 represent investor preferences and are given before solving the problem. Hence they are parameters.The values
of the goal variables $d_i, i = 1, 2, 3, 4$ are supposed to measure the amount by which our selection process underachieves with respect to the best scenario for each problem.

Note that here we used the elegant matrix notation for skewness and kurtosis of the portfolio, but the paper proposes a different formula that avoid the Kronecker products and use instead
the sample definitions, but it is also necessarily much longer and much more cumbersome to write down. Practitioners will probably have to resort to use those, but here we wanted to stress
the methodology more than anything else.

There is one more way to approach the problem, one that amounts to a true multicriteria decision analysis (MCDA). This is more complicated and it leads to multiobjective programming in
which more than one objective functions are optimized simultaneously. A multiobjective optimization looks like this:

$$\begin{aligned}&\min_{w} \{f_1(w), f_2(w), \ldots, f_r(w)\}\\&\text{subject to } \; Aw \leq a\\&\qquad\qquad w \geq 0\end{aligned}$$

where $f_1(w) = w^T M_1$, $f_2(w) = w^T M_2 w,$ etc. In multiobjective programming, we do not have an optimal solution as it is unfeasible to optimize
all objective functions simultaneously. For this reason, the concept of optimal solution is replaced by the concept of {\bf Pareto optimal} solution.
A feasible solution w has the property to be {\bf Pareto dominant}, i.e., if v is another feasible solution, then $f_i(w) \leq f_i(v)$ for $i = 1, 2, \ldots, r$ and $f_i(w) < f_i(v)$ for at least one value of $i = 1, \ldots, r.$

8. Conclusion

This lesson introduced a more general way to perform portfolio selection by taking into consideration skewness and kurtosis of the resulting portfolios directly. In addition, it also showed how to account for liquidity. Finally, we discussed an example of a

true multi-objective optimization problem, which is a much more difficult problem, and we will see a bit more of it in Module 3, Lesson 4.

In the next lesson, we will study how to deal with currency management, an issue that is of importance when a portfolio does not consist of assets all denominated in the same domestic currency.

References

- Kim T., Does individual-stock skewness/coskewness reflect portfolio risk? Finance Research Letters. Volume 15, November 2015, p. 167-174
- M. Mhiri, JL Prigent, International Portoflio Optimization with Higher Momentsm International Journal of Economics and Finance, Vol 2. No 5, November 2010

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Lesson Notes

MODULE 2 | LESSON 3

OPTIMAL PORTFOLIOS AND CURRENCY MANAGEMENT

Reading Time	120 minutes
Prior Knowledge	Portfolio optimization, Financial instruments like forward contracts, options (vanilla and exotic), etc.
Keywords	Currency risk, Calculating portfolio return in domestic currency, Risk of portfolio in domestic currency, Bid/ask spread for currencies, Interest rate parity, Purchasing power parity, Passive hedging, Discretionary hedging, Active hedging

In the previous lesson we look at how to optimize portfolios based on more than just mean and variance and introduced skewness, kurtosis, and liquidity. In this lesson we add an additional element of complications: foreign assets and currency risk.

1. Generalities on Working with Currencies

A well-diversified portfolio is likely to consist of domestic assets (denominated in the investor's home currency) and foreign assets (denominated in a currency different from the home currency). This creates additional complications for both risk and returns. At a minimum, leaving aside specific country risks, when an investment is priced in a currency different from the domestic one, there are two sources of risk:

- the return on the foreign asset denominated in the foreign currency;
- the return on the foreign currency, which derives from the change in the exchange rate between the domestic and the foreign currency.

Let r_D and r_F represent the return from investing in an asset denominated in domestic and foreign currency, respectively, and let r_{FX} be the rate of change in the foreign currency vs. the domestic currency. Then, we have

$$r_D = (1 + r_F)(1 + r_{FX}) - 1 \approx r_F + r_{FX}.$$

In some instances, r_{FX} could be known, but often it is not. This creates a need to hedge the global (domestic+foreign) portfolio. If we decided that we want to hedge the currency exposure, we would need to sell short the ending amount of foreign currency units, which is not known in the case of risky assets and makes managing currency risk complicated.

Example (Calculate returns in domestic currency): Consider an Italian investor (home currency is EUR) who invests in a portfolio of natural resources that trade in Canadian dollars (foreign currency is CAD). Over a one-year period, the value of the investor's portfolio has increased by 10% and the CAD-EUR exchange rate went from 1.30 to 1.20. What is the investor's return in her domestic currency?

Note that the exchange rate went from 1.3 CAD for 1 EUR to 1.2 CAD for 1 EUR, i.e., the euro has depreciated vs. the Canadian dollar, which is good news for the investor as she gets paid in the foreign currency. More specifically, we went from 1.3 CAD for 1 EUR to 1.2 CAD for 1 EUR or .769 EUR for 1 CAD to 0.833 EUR for 1 CAD, i.e., the foreign currency had a return of 8.32% vs. the domestic one. Then, using the formula above, we find that

$$r_{EUR} = (1 + 10\%)(1 + 8.32\%) - 1 = 19.2\%.$$

Using the approximate formula, we would find

$$r_{EUR} = 10\% + 8.32\% = 18.32\%$$

so the formula is not always accurate unless the returns on both the assets and the currencies are small.

In the case just depicted, the exchange rate moves in the investor's favor, but if the exchange rate had gone from 1.30 to 1.40 instead, then the EUR would have appreciated vs. the Canadian dollars (bad news for the investor who is paid in Canadian dollars).

In this case we go from receiving 0.769 EUR for each CAD to receiving 0.714 EUR for each CAD, a return on the foreign currency of -7.12%. Hence:

$$r_{EUR} = (1 + 10\%)(1 - 7.12\%) - 1 = 2.17\%,$$

a dramatic difference when compared to the previous case.

While pure quants are not usually tasked with managing portfolios and currency risk, they may have to support a portfolio manager, and nothing stops them from potentially becoming portfolio managers if they so desire. Therefore, it is important to understand currency risk. For this reason, we will devote the remainder of the lesson to understanding currency risk and how to manage it.

2. Basics of Exchange Rates

To deal with exchange rate movements, one must understand what causes exchange rates to move in the first place. We will not devote a lecture to International Economics, but we will summarize the key issues so that a student can understand and be competent in the essential elements of exchange rates.

We start with some terminology. The base currency is the denominator of the exchange rate, and it is priced in terms of the currency in the numerator.

So, for instance, if we quote CAD/USD = 1.25, we mean that 1 USD is worth 1.25 CAD.

We will assume that the base currency is the domestic currency for what we do in this lesson. This has the merit of establishing a clear way to quote currencies vs. other currencies. In general, currencies are quoted as bid/ask spreads. For example:

	bid	ask
Spot CAD/USD	1.2570	1.2580

How do we read this table for practical purposes? Say that a U.S. investor has USD $1,000,000 to convert to CAD. Then, the investor is buying CAD and selling USD. Therefore, the bid price is the relevant one, and the investor will get CAD 1,257,000. Now, assume that the same investor receives CAD 1,500,000 from an investment in Canada and needs to convert the amount to USD. In this case, the investor is selling CAD and buying USD, and the relevant price is the ask price. Then, the investor will get USD $1,500,000/1.2580= 1,192,368.84. Often, the easiest way to remember how it works is to think that one always gets the worst of the two prices for what they need to buy/sell!

Currency quotes can also be defined as **direct** or **indirect**. An indirect quote is when the foreign exchange rate is reported in terms of units of foreign currency per unit of domestic currency. This is what we have done above. A direct quote is when currency exchange is expressed in terms of units of domestic currency per unit of foreign currency. The relationship between the two is very simple:

$$\text{direct quote} = \frac{1}{\text{indirect quote}} \quad \text{and} \quad \text{indirect quote} = \frac{1}{\text{direct quote}}$$

Make sure that you are aware of which one you are given while reading quotes or scholarly papers.

Now that we know how to read currency quotes, we ask ourselves the following question: What moves the exchange rates between two currencies? We will consider the two main models.

2.1 Interest Rate Parity Model

This model proposes a relationship between the interest rates of two given currencies and the spot and forward exchange rates between the currencies. As such, it can be used to predict how exchange rates between two currencies change when the risk-free interest rates of the two currencies are known. This theory is based on the assumption that no arbitrages should exist, not for long at least, and that is also why it is considered a model for the short/medium period. The forward exchange rate is the exchange rate at which a party is willing to enter into a contract to receive or deliver a currency at some future date. Essentially, this is what matters as one makes certain investments to receive future benefits or as a manufacturer may purchase items to be delivered and paid for in the future. Let S be the spot rate (the exchange rate right now) and let F be the forward rate (the exchange rate in the future). Furthermore, let r_d be the interest free rate of return on deposits in the domestic currency and r_f the interest free rate of return on deposits in the foreign currency. Then, if n stands for the number of years, the formula that links F and S is the following:

$$F = S \cdot \left(\frac{1 + r_f}{1 + r_d} \right)^n.$$

Example: Let's assume that the current exchange rate between Japanese Yen (JPY) and USD is 115.70. Also, assume that the interest rate in Japan is 4% and the interest rate in the U.S. is 2%. What do we expect will happen to the forward exchange rate between the two currencies in a one-year time? Well, $S = 115.7, r_f = 4\%, r_d = 2\%, n = 1$ and using the formula given earlier:

$$F = 115.7 \times \left(\frac{1+4\%}{1+2\%}\right)^1 = 117.97,$$

i.e., the Yen is depreciating. This is natural because, otherwise, there would be an arbitrage opportunity. For instance, assume that the forward exchange rate were 110. A U.S. investor could walk to a bank in the U.S. and borrow USD $1,000,000 for 2% interest rate. Then, the investor can convert the money to JPY at the spot rate. Therefore, the investor receives YEN 1,000,000 x 115.70 = 115,700,000, which can be invested in the foreign country at 4%. After one year, the U.S. investor receives JPY 115,700,000 x 1.04 = 120,328,000. At this point, the investor converts back to USD using the forward rate, i.e., the investor gets USD 120,328,000/110 = 1,093,890.91. Now the investor pays the interest rate on the USD $1,000,000 borrowed from the bank, i.e., USD $20,000 and has netted a profit of USD $73,890.91. Hence, the only way to avoid arbitrage is if JPY depreciates.

2.2 Purchasing Power Parity Model (PPP)

In this case, the exchange rate of any two currencies will remain equal to the ratio of their respective purchasing powers. As usual, the purchasing power of a currency is measured as the amount of the currency needed to buy a selected product or basket of goods commonly available in different countries, and it relates to inflation. So, if we define by i_f and i_d the inflation rate in the foreign and domestic country, respectively, we have the following formula:

$$F = S \cdot \left(\frac{1+i_f}{1+i_d}\right)^n.$$

Example: Assume that the exchange rate GBP/USD = 0.75. Also, assume that the price of a standardized basket of goods is $20,000 in the U.S. and GBP 15,000 in the UK. Furthermore, we believe that in one year, the same basket will cost $22,000 in the US and GBP 16,000 in the UK. Hence, $i_d = i_US = 22,000/20,000 - 1 = 10$ and $i_f = i_UK = 16,000/15,000 - 1 = 6.67$ Then,

$$F = 0.75 \cdot \left(\frac{1+6.67\%}{1+10\%}\right)^1 = 0.7273,$$

i.e., GBP will appreciate against USD.

3. Calculating Portfolio Return for Investments in Foreign Assets

If we assume that an investor owns assets across multiple markets with different currencies, the first task is to understand how to compute the return on such a portfolio. As the return is a weighted average of returns, we need to how compute weights and returns. It seems only logical that:

- the weights are computed as the percentage of the portfolio invested in each foreign currency denominated asset expressed in terms of domestic currency.
- the returns are also expressed in terms of the domestic currency.

Thus,

$$r_p^{(d)} = \sum_{j=1}^{m} w_j r_j^{(d)},$$

where each index value of j refers to a different currency. There is no problem in aggregating all assets in the same currency as return is a linear function.

Example: Suppose that we have an investor based in Canada and that this investor has GBP 1,000,000 invested in UK assets, EUR 2,000,000 invested in Germany, and USD $3,000,000 invested in US assets.

We are also given that $r_{US} = 10\%, r_{EUR} = 8\%,$ and $r_{GBP} = 7\%.$ In addition we are also told that

$$\textbf{beginning CAD/GBP} = 1.50 \quad \textbf{ending CAD/GBP} = 1.55;$$

$$\textbf{beginning USD/CAD} = 0.75 \quad \textbf{ending USD/CAD} = 0.78;$$

$$\textbf{beginning EUR/CAD} = 0.70 \quad \textbf{ending EUR/CAD} = 0.68.$$

Task 1. Translate all assets in domestic (CAD) currency at the beginning of the period:

$$A^{GBP} = 1,000,000 \cdot (1.50) = CAD1,500,000;$$

$$A^{EUR} = 2,000,000 \cdot (1/0.70) = CAD2,857,143;$$

$$A^{USD} = 3,000,000 \cdot (1/0.75) = CAD4,000,000.$$

Task 2. The total amount invested in the international portfolio is $A^{CAD} = 8,357,143.$ This allows us to compute the weights:

$$w_{GBP} = 1,500,000/8,357,143 = 0.179, \quad w_{EUR} = 2857,143/8,357,143 = 0.342,$$

$$w_{USD} = 4,000,000/8,357,143 = 0.479.$$

Task 3. Compute the returns for each class of assets in domestic currency (CAD):

$$r_{GBP}^{CAD} = (1+7\%)(1+(1.55/1.50-1)) - 1 = 1.07 \cdot 1.0333 - 1 = 10.56\%;$$

$$r_{EUR}^{CAD} = (1+8\%)(1+((1/0.68)/(1/0.70)-1)) - 1 = 1.08 \cdot 1.0294 - 1 = 11.18\%$$

$$r_{USD}^{CAD} = (1+10\%)(1+((1/0.78)/(1/0.75)-1)) - 1 = 1.10 \cdot 0.9615 - 1 = 5.77\%$$

Note that the CAD appreciated against the USD and depreciated against both GBP and EUR.

Task 4. Compute the return in domestic currency:

$$r_P^{CAD} = 0.179 * 10.56\% + 0.342 * 11.18\% + 0.479 * 5.77\% = 8.48\%.$$

4. Portfolio Risk for Investments in Foreign Assets

Investments involving foreign denominated assets have two sources of risk: the change in price of the investments (in the foreign currency) and the fluctuations in the exchange rate between the

foreign currencies and the domestic one.

The first task awaiting us is to define the volatility of a foreign denominated asset in the international portfolio. For each foreign denominated asset j, using the approximation

$$r_j^{(d)} \approx r_j^{(f)} + r_j^{(fx)}$$

we derive the following formula for the variance:

$$\sigma^2(r_j^{(d)}) \approx \sigma^2(r_j^{(f)}) + \sigma^2(r_j^{(fx)}) + 2\sigma(r_j^{(f)})\sigma(r_j^{(fx)})\rho(r_j^{(f)}, r_j^{(fx)})$$

where $\rho(r_j^{(f)}, r_j^{(fx)})$ is the correlation for the return of the asset j in the foreign currency and the fluctuations of the currency of asset j itself.

Similarly, $\sigma(r_j^{(fx)})$ is used to denote the volatility of the currency for asset j. The notation is arguably cumbersome, but it shows that the variance for the return of asset j in the domestic currency is higher than the return of asset j in its foreign currency.

However, if the correlation for the return of the asset j in the foreign currency and the fluctuations of the currency of asset j is negative, the volatility of the returns in the domestic currency is dampened by it.

Now, if we use the formula for $\sigma^2(r_j^{(d)})$ just introduced, we can define the variance for the portfolio as

$$\begin{aligned}\sigma_P^{2,(d)} &= \sigma^2\left(\sum_{j=1}^{m} w_j r_j^{(d)}\right) \\ &= \sum_{j=1}^{m} w_j^2 \sigma^2(r_j^{(d)}) + 2\sum_{j=1}^{m}\sum_{h>j}^{m} w_j w_h \sigma(r_j^{(d)})\sigma(r_h^{(d)})\rho(r_j^{(d)}, r_h^{(d)}).\end{aligned}$$

To complete this section, we need one last effort. One notices the term $\rho(r_j^{(d)}, r_h^{(d)})$, and we need to write it in terms of $r_j^{(f)}$ and $r_h^{(f)}$ and the exchange rates. Recalling the definition of correlation, the denominator is not a problem as we already have $\sigma^2(r_j^{(d)})$ and $\sigma^2(r_h^{(d)})$, and can concentrate on the covariance term in the numerator.

Thus, using the bilinearity property of the covariance operator, we find

$$\sigma(r_j^{(d)}, r_h^{(d)}) = \sigma(r_j^{(f)}, r_h^{(f)}) + \sigma(r_j^{(f)}, r_h^{(fx)}) + \sigma(r_h^{(f)}, r_j^{(fx)}) + \sigma(r_j^{(fx)}, r_h^{(fx)}).$$

Therefore,

$$\begin{aligned}\rho(r_j^{(d)}, r_h^{(d)}) &= \frac{\sigma(r_j^{(d)}, r_h^{(d)})}{\sigma(r_j^{(d)})\sigma(r_h^{(d)})} \\ &= \frac{\sigma(r_j^{(f)}, r_h^{(f)}) + \sigma(r_j^{(f)}, r_h^{(fx)}) + \sigma(r_h^{(f)}, r_j^{(fx)}) + \sigma(r_j^{(fx)}, r_h^{(fx)})}{\sigma(r_j^{(d)})\sigma(r_h^{(d)})}\end{aligned}$$

with $\sigma(r_j^{(d)})$ and $\sigma(r_h^{(d)})$ defined above.

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5. Optimal Portfolio with Multiple Currencies

Not unlike what we saw in previous lessons, optimization in the case of a portfolio that involves multiple currencies requires that one selects weights that identify a portfolio on the efficient frontier for the best trade-off between mean and variance. To achieve this, we need to estimate expected returns and volatilities of each of the foreign currency denominated assets. This adds a lot of parameters to our model as we would need the returns of each asset in the foreign currency, $r_j^{(f)}$ for $j = 1, \ldots, m$, the returns on the foreign currency of each of the assets, $r_{j,fx}$, volatility of foreign denominated returns, $\sigma(r_j^{(f)})$, volatility on the exchange rates $\sigma(r_{j,fx})$, as well as many correlations.

However, if we could do that, then the problem would be no different than what we have seen already. In its simplest version:

$$\begin{aligned} &\max_w w^T r^{(d)} - \tfrac{1}{2} w^T \Sigma^{(d)} w \\ &\text{subject to} \quad w^T \iota = 1 \end{aligned}$$

where $r^{(d)}$ and $\Sigma^{(d)}$ represent the parameters expressed in domestic currency and that were defined in the previous section.

Estimating $r^{(d)}$ is not a big deal computationally (although estimates could be unreliable); estimating $\Sigma^{(d)}$ is cumbersome.

An exact and computable solution to the general quadratic mean-variance problem that incorporates exchange rate and return volatility can be obtained following an approach by Rustem (1995). We present the relevant steps:

Step 1. Let r^i represent the return vector on n^i investments in currency i and assume that

$$r^i = E[r^i] + \rho^i, \;\; \rho_i \sim N(0, \Lambda^i_{\rho\rho}), i = 1, 2, \ldots, I$$

where $\Lambda^i_{\rho\rho}$ is the $n^i \times n^i$ variance covariance matrix for returns in currency i.

Step 2. Let e^i be the exchange rate associated with currency i, i.e.

$$e^i = E[e^i] = \epsilon^i \;\; \epsilon_i \sim N(0, \Lambda^i_{\epsilon\epsilon}), i = 1, 2, \ldots, I$$

where $\Lambda^i_{\epsilon\epsilon}$ is the $n^i \times n^i$ variance covariance matrix for returns in currency i.
The exchange rate corresponding to the base currency would be equal to 1 and its variance-covariance matrix is the null matrix.

Step 3. Let r and ρ denote the $\sum_{i=1}^{I} n^i \times 1$ vectors of returns on all assets and the corresponding uncertainty, that is

$$r = E[r] + \rho.$$

Step 4. Let e by the $I \times 1$ vector of all exchange rates, and ϵ their uncertainty, i.e.,

$$e = E[e] + \epsilon.$$

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Step 5. Now let

$$\beta = \begin{pmatrix} \rho \\ \epsilon \end{pmatrix} \sim N(0, \Lambda)$$

with

$$\Lambda = \begin{bmatrix} \Lambda_{\rho\rho} & \Lambda_{\rho\epsilon} \\ \Lambda_{\epsilon\rho} & \Lambda_{\epsilon\epsilon} \end{bmatrix}.$$

An estimate for $\Lambda_{\rho\rho}$ can be obtained from historical returns data; for $\Lambda_{\epsilon\epsilon}$, we can use the error between past
forecasts and historical exchange rates. Finally,$\Lambda_{\rho\epsilon}$ can be obtained using historical returns and forecast errors.

Step 6. This is the key step to be able to write the problem in a convenient matrix form. To this purpose, we introduce a matrix q whose dimensions are $\sum_{i=1}^{I} n^i \times I$ **:** defined as follows:

$$q_{ij} = \begin{cases} 1 & \textbf{if investment i is in currency j} \\ 0 & \textbf{otherwise.} \end{cases}$$

Note that if we have one investment for each currency, then $Q = I$, the unit matrix.

Example: Suppose we have stocks in three different currencies, A, B, and C. Two stocks are denominated in currency A, 1 in currency B, and 2 in currency C. What is the q matrix in this case? In this case,$n^1 = 2, n^2 = 1, n^3 = 2.$ Therefore,q is a5×3 matrix that looks like this:

$$Q = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{pmatrix}$$

Step 7. Introduce one more matrix:

$$P = \mathrm{diag}(Qe)$$

which is of dimension $\sum_{i=1}^{I} n^i \times \sum_{i=1}^{I} n^i$ and whose diagonal i-th diagonal element is given by the i-th element of $Qe.$

Example: Using the setup of the previous example, if $e^A, e^B,$ and e^C denote the exchange rates of the foreign currencies vs. the domestic one, we have

$$Qe = \begin{pmatrix} e^A \\ e^A \\ e^B \\ e^C \\ e^c \end{pmatrix} \quad \text{and} \quad P = \begin{pmatrix} e^A & 0 & 0 & 0 & 0 \\ 0 & e^A & 0 & 0 & 0 \\ 0 & 0 & e^B & 0 & 0 \\ 0 & 0 & 0 & e^C & 0 \\ 0 & 0 & 0 & 0 & e^C \end{pmatrix}$$

Step 8. Equipped with all this, the return in the base currency can be written as

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$$R = Pr$$

which is a $\sum_{i=1}^{I} n^i \times 1$ vector.

Step 9. Let w be the $\sum_{i=1}^{I} n^i \times 1$ vector of portfolio weights for the multi-currency assets. Then, the return for the portfolio can be written as

$$w^T R = r^T(\text{diag}(w)P)\iota.$$

Step 10. Now from a theorem proved (this proof is not hard) by Rustem, we can write:

$$E[w^T R] = (E[r])^T(\text{diag}(w)PE[e]) + \frac{1}{2}\text{tr}(\Lambda\Omega(w))$$

where $\Omega(w)$ is the following matrix

$$\Omega(w) = \begin{bmatrix} 0 & \text{diag}(w)Q \\ Q^T\text{diag}(w) & 0 \end{bmatrix}.$$

Step 11. Now we get the variance for the returns in the domestic currency using a second theorem proved by Rustem (this is harder):

$$\text{Var}(w^T R) = w^T M^T \Lambda M w + \frac{1}{2}\text{tr}(\Lambda\Omega(w))^2$$

with

$$M = \begin{bmatrix} \text{diag}(QE[e]) \\ Q^T\text{diag}(E[r]) \end{bmatrix}$$

an $(\sum_{i=1}^{I} n^i + I) \times \sum_{i=1}^{I} n^i$ matrix whose first $\sum_{i=1}^{i} n^i$ rows consists of the diagonal matrix whose diagonal elements are taken from the matrix $QE[e]$, and the following I elements are the byproduct of q with a diagonal matrix whose diagonal elements are $E[r]$.

Through the ten steps above, we are now in a position to write the QP to find the optimal portfolio:

$$\begin{gathered} \max_w \ (E[r])^T(\text{diag}(w)PE[e]) + \tfrac{1}{2}\text{tr}(\Lambda\Omega(w)) - \tfrac{1}{2}\left(w^T M^T \Lambda M w + \tfrac{1}{2}\text{tr}(\Lambda\Omega(w))^2\right) \\ \text{subject to} \ \ w^t\iota = 1. \end{gathered}$$

Note that while this is a QP, it is not a standard one. Standard QP look like this (just a reminder as we saw this before):

$$\begin{aligned} \text{minimize} \quad & (1/2)x^T Px + q^T x \\ \text{subject to} \quad & Gx \leq h \\ & Ax = b \end{aligned}$$

6. Strategic Decisions for Portfolio Optimization

In a world where information is complete and freely available, it would be desirable to optimize the portfolio with respect to all these exposures, considering each currency simultaneously.

Unfortunately, in the real world, this is a high task and portfolio managers who are forced to deal with these situations have developed two main approaches:

- fully hedge the portfolio for currency exposure;
- decide which currency they would like to be exposed and hedge the others.

This is possible because the use of derivatives allows us to separate different risk exposures from the other and can manage the price risk (assets selection) separately from exchange-rate risk. Ideally, if there existed a perfect hedge that were free or at least achievable at reasonable costs, currency exposure could be neutralized and, in that case, $\sigma^2(r_j^{(d)}) = \sigma^2(r_j^{(f)})$, our optimization process would be much simpler.

Neither academic research nor empirical analysis produce a strong conclusion about how currency risk is best managed. Among the reasons not to hedge we find:

- hedging is expensive and requires a lot of time;
- in the long run, currencies become a zero-sum game in which if a currency changes against the portfolio, another is likely to change in our favor;
- over time, currencies tend to revert to their (theoretical) fair value and movements away from that value are only temporary.

Practitioners have resorted to three main strategies:

- **Passive hedging:** In this case, portfolio managers limit themselves to matching the currency exposure of the portfolio to that of the benchmark used to evaluate their performance. It does not mean that the portfolio manager does nothing, but rather, their management is limited to periodic re-balancing to keep the exposure in line with that of the benchmark.
- **Discretionary hedging:** This is very similar to passive hedging, except that the portfolio manager is allowed to deviate from the currency's exposure in the benchmark. For example, the manager could be allowed to deviate 5-10% from the exposure in the benchmark.
- **Active management:** This is similar to discretionary hedging, but the manager is allowed more discretion and is expected to generate incremental returns (hopefully positive!)

The essence of the three approaches could be summarized as follows:

- passive hedging = eliminate currency risk relative to benchmark;
- discretionary hedging = reduce currency risk and pursue modest incremental returns;
- active management = generate additional alpha; reducing risk is secondary or not required.

It is not unusual for active management to be outsourced to an external party that specializes in currency management (**currency overlay**).

Regarding active management let's ask ourselves: what happens if we were to hedge everything? To this purpose, consider a simple scenario: We own one foreign asset and we buy currency forward contracts. In this case, the variance of the hedged porttfolio (1 foreign asset, (a), and 1 forward currency contract, (f)) can be written as

$$\sigma_p^2 = w_a^2\sigma_a^2 + w_f^2\sigma_f^2 + 2w_a w_f \sigma_a \sigma_f \rho(a, f),$$

where the symbols have the usual meaning and $\rho(a, f)$ represents the correlation between the foreign asset and the forward currency contract.

If we assign a weight of 1 to the asset, then the previous equation becomes

$$\sigma_p^2 = \sigma_a^2 + w_f^2\sigma_f^2 + 2w_f\sigma_a\sigma_f\rho(a, f),$$

which is ideal for calculating the optimal weight for the forward contract in order to minimize the variance of the portfolio. Simple calculus yields

$$w_f = -\frac{\sigma_a}{\sigma_f}\rho(a, f).$$

This tells us two things. First, it tells us that a higher correlation between asset and currency indicates that the exposure to currency increases the risk (less diversification) and requires a higher minimum-variance hedge ratio. In addition, it also tells us that a higher σ_f (the standard deviation of the currency) decreases the hedge.

Example (Hedging 100% of currency exposure is not always optimal): Assume that we have a portfolio consisting of one foreign denominated asset that has a10% standard deviation. Assume also that the foreign currency has a standard deviation of $15\%,$ and the correlation between the asset and the currency is $0.6.$ Then, according to the formula we just derived, the minimum-variance hedge ratio is equal to $-0.10/0.15 \cdot 0.6 = -0.4.$ In this case, the standard deviation of the hedged portfolio would be

$$\sigma_p = \sqrt{0.10^2 + (-0.4)^2 \cdot 0.15^2 + 2 \cdot (-0.40) \cdot 0.10 \cdot 0.15 \cdot 0.6} = 0.080.$$

Hence, the hedge helped to lower the risk of the portfolio. Now, we choose to hedge 100 of the position, i.e.,$w_f = -1.$ Then, using the same formula yields:

$$\sigma_p = \sqrt{0.10^2 + (-1)^2 \cdot 0.15^2 + 2 \cdot (-1) \cdot 0.10 \cdot 0.15 \cdot 0.6} = 0.1204.$$

In other words, a 100 hedge ratio is not the ratio that minimizes portfolio risk, and, in this example, a fully hedged portfolio has a higher volatility than the unhedged one!

Among active hedging strategies, we could list what is often called the **minimum regret hedge ratio**. In this case, investors hedge 50% of their currency exposure without any consideration of volatilities and correlations. The philosophy behind this approach should be clear: Currency exposure can be a friend (if it moves the investor's way) or a foe (it it moves the other way), and therefore. an investor who is prone to suffer from regret should choose between 0% hedge or 100% hedge, but does not know which one is the right course of action. Going for 50% minimizes the investor's *ex post regret*. This approach has been criticized because it does not take into consideration either the probability distribution of possible outcomes or the change in utility when the hedging strategy outperforms or underperforms. In addition, there is a lack of valid empirical studies to test this particular strategy.

7. Hedging Currency Exposure: Implementation

We need to distinguish between two large categories of hedging: linear and non-linear strategies. A linear strategy is one where the return of the hedged portfolio is a linear function of the return on the hedged currencies. Typically, a linear hedge neutralizes both potential losses and potential gains. A non-linear hedging strategy, on the other hand, protects against losses but preserves the potential upsides in the portfolios due to currency exposure and usually refers to using options.
Among the linear strategies, we can list approaches like un-hedged, full-hedged, currency-specific hedging, and more.

There is also a choice among a strategy where the hedge is fixed (**static hedge**) and one in which the hedge changes over time (**dynamic hedge**). In the latter case, we could possibly need to perform heavy re-balancing.

We should look now at how active hedging is implemented with forward contracts. Forward contracts are usually the preferred hedging vehicle since forwards can be customized, forwards are available for almost any currency and in any size, and trading volume is very high.

Say that we have a long exposure to Canadian dollars (CAD) that we would like to hedge. If we expect that the CAD will appreciate, we can reduce our hedge to be hedging less than full exposure (never lose sight that hedging is expensive and not free). On the other hand, if we expect CAD to depreciate, we would increase our hedge. This strategy creates **positive convexity** (increase gains and reduce losses). As far as implementing the hedging, we could use one of the following strategies:

- Buy a put option at the money (portfolio insurance). In this case, we have high initial costs, but no opportunity costs.
- Same strategy but using out-of-the-money puts. This is less expensive, but there is downside up to the strike price of the puts.
- Cheaper still would be to use a collar: For instance, we can buy 35-delta puts on the CAD and sell 35-delta calls on the CAD. This gives us downside protection while removing some upside potential, but it reduces the cost of hedging.
- A put spread where we buy a 35-delta put and sell a 25-delta put. The downside protection does not begin until the strike of the puts we purchased, it is relatively cheap to set up, but if the CAD falls below the strike of the 25-delta put, all the protection is gone.
- A seagull spread is achieved by going long a 35-delta put and a 25-delta put and selling a 35-delta call. This lowers the cost of the hedging strategy even more and provides the same protection as a put spread, but we give up upside potential.
- Knock-in/knock-out options, which provide the same protection/upside of plain vanilla options but come alive only if the CAD falls below a certain threshold or ceases to exist if the CAD surges above a specific threshold (no need for protection anymore).

The table below summarizes a few key facts about hedging currencies.

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ELEMENT	EVENT	ACTION
Currency price	Appreciate	Reduce hedge
	Depreciate	Increase hedge
Volatility	Rises	Long straddle, strangle
	Decreases	Short straddle, strangle
Economy/Market	Stable	Carry trade
	Deteriorates	Stop carry trade

The performance of option strategies can be evaluated via Monte Carlo simulations or back-testing. However, in line with what we have seen in this module so far, we should keep in mind that back testing relies only on one specific outcome (what occurred, i.e., one of possibly infinite outcomes), and Monte Carlo simulations allows us to test our strategy against a large number of possible outcomes.

8. Conclusion

In this lesson, we saw that when a portfolio contains assets denominated in foreign currencies, there are a few complications that one has to deal with. We looked at how to properly read exchange rates, how to calculate their impact on portfolio returns and volatility, and what changes if the exposure to currency is hedged or not hedged. Finally, we considered a few hedging strategies. Understanding these nuances is important to knowing how to work with problems involving foreign assets and/or assets denominated in foreign currencies.

In the next lesson, we will look at factor models and what their implications are, both in terms of reducing the parameters to estimate and as a tool in and of themselves.

References

- Rustem, B. "Computing Optimal Multi-Currency Mean-Variance Portfolios." *Journal of Economics Dynamics and Control*, vol. 19, 1995, pp. _901–908, _ https://doi.org/10.1016/0165-1889(94)00812-V

Lesson Notes

MODULE 2 | LESSON 4

FACTOR MODELS IN PORTFOLIO THEORY

Reading Time	120 minutes
Prior Knowledge	Markowitz mean-variance portfolio optimization, regression models using both ordinary least squares and weighted or generalized least squares, multi-step estimation methods.
Keywords	Macroeconomic factors, fundamental factors, statistical (latent) factors, factor-mimicking portfolio, return-based style analysis

In the previous lesson, we looked at what changes are necessary when a portfolio contains assets denominated in foreign currencies and whether one should hedge one's exposure via derivatives. In this lesson, the emphasis shifts back to the original problem of min-variance optimization. Estimating all the parameters necessary is hard, often impossible, and without parameters, we cannot perform any optimization in a reliable manner. Factor models allow us to reduce the parameters to be estimated considerably. And, factors can have a life of their own and we will look at some of the possible applications.

1. Generalities

By introducing structural multi-factor risk models, it is possible to shift Markowitz's approach to portfolio construction toward a two-stage approach.

In the first stage, individual assets are grouped into classes and the optimization is performed at the level of asset classes. Once asset allocation has been completed, managers specializing in selecting securities step in.

While asset classes are usually small (20–30, occasionally more), the selection of individual securities can require the analysis of thousands of individual items. When there are so many units to analyze, it becomes impossible to estimate all the parameters one needs from the data.

This problem led to the introduction of structural multi-factor risk models, which attempt to identify a certain (relatively small) number of common factors that are able to explain the returns of individual securities.

Factors and asset classes are different. Asset classes are usually well-defined, stable, and directly investable. On the other hand, factors represent exposure to risk and, in general, are not directly investable. Some of them are even latent (hidden). The process of identifying and measuring factors is both art and science. We can distinguish between three types of factors:

- observable economic/financial time series (**macroeconomic factors**);
- created from observable asset characteristics (**fundamental factors**);
- not observable and extracted from asset returns (**statistical or latent factors**).

The first two groups can comprise economic variables like inflation, GNP growth, accounting ratios like P/BV (price-to-book value), EPS or EPS growth, etc. On the other hand, the third group contains those factors that are like principal components and that may or may not have an obvious physical meaning.

To stress the point about the difference between asset classes and factors, one should keep in mind that factors appear to have strong diversification properties because they often have low correlations with each other and sometimes even negative correlations with one another. On the other hand, asset classes usually exhibit strong positive correlations.

One question will have to be answered: whether factors really improve diversification or not. Nevertheless, by constructing portfolios based on risk factors, it is possible to remove what are sometimes artificial constraints in the definition of asset classes and managers can focus on risk drivers that affect their entire portfolio.

2. Factor Models Setup

This section is important for becoming acquainted with the necessary statistical language, which will later allow us to estimate loading parameters and factors themselves. Also, there are some important nuances that need to be well understood.

The three types of factors listed above can be treated just like the same object. We can specify factor models in three different ways.

The general form is:

$$R_{it} = \alpha_i + \sum_{k=1}^{K} \beta_{ki} f_{kt} + \epsilon_{it} = \alpha_i + \beta_i^T f_t + \epsilon_{it}$$

where:

- R_{it} is the return on asset i for $i = 1, 2, \ldots, M$ in time period $t = 1, 2, \ldots, T$. The returns can be real returns if there are no risk-free assets and excess returns when a risk-free asset is available;
- f_{kt} is the k-th factor $k = 1, 2, \ldots K$ at time t;
- β_{ik} is the sensitivity beta (also called loading) for asset i with respect to factor k;
- ϵ_{it} is the asset specific (residual factor).

Some assumptions are needed:

- the factors need to be stationary and their unconditional moments are given by

$$E[f_t] = \mu_f, \;\; \text{cov}(f_t) = \Sigma_f.$$

- the error terms are uncorrelated with each one of the common factors, i.e.,

$$\text{cov}(f_{kt}, \epsilon_{it}) = 0 \;\; k = 1, 2, \ldots, K, i = 1, 2, \ldots, M, \quad \text{and} \quad t = 1, 2, \ldots, T.$$

- Finally, the error terms are serially uncorrelated, but contemporaneously correlated across assets, that is

$$\text{cov}(\epsilon_{it}, \epsilon_{js}) = \begin{cases} \sigma_i^2 & \text{for } i = j, t = s; \\ 0 & \text{otherwise .} \end{cases}$$

Furthermore, we will use the convention that

$$R_t = \begin{pmatrix} R_{1t} \\ R_{2t} \\ \vdots \\ R_{Mt} \end{pmatrix} \;\; t = 1, 2, \ldots, T$$

represents a vector that is the **cross section** of all assets' returns at time t.

On the other hand, if we wrote

$$R_i = \begin{pmatrix} R_{i1} \\ R_{i2} \\ \vdots \\ R_{iT} \end{pmatrix} \;\; i = 1, 2, \ldots, M$$

that should be recognized as the **time series** of the returns of asset i over the time framework used.

Finally, all assets' returns over the full time period are represented as the matrix

$$R = \begin{pmatrix} R_{11} & R_{21} & \cdots & R_{M1} \\ R_{12} & R_{22} & \cdots & R_{M2} \\ \vdots & \vdots & \ddots & \vdots \\ R_{1T} & R_{2T} & \cdots & R_{MT} \end{pmatrix}.$$

Hence, we examine factor models through 3 different lenses:

- time series regressions;
- cross-section regressions;
- multivariate regressions.

As a **cross-section regression**, this model looks like:

$$R_t = \alpha + Bf_t + \epsilon_t \;\; t = 1, 2, \ldots, T.$$

In this setup:

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$$R_t = \begin{pmatrix} R_{1t} \\ R_{2t} \\ \vdots \\ R_{Mt} \end{pmatrix}, \ \alpha = \begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_M \end{pmatrix}, \ \epsilon_t = \begin{pmatrix} \epsilon_{1t} \\ \epsilon_{2t} \\ \vdots \\ \epsilon_{Mt} \end{pmatrix},$$

$$f_t = \begin{pmatrix} f_{1t} \\ f_{2t} \\ \vdots \\ f_{Kt} \end{pmatrix}, \ B = \begin{pmatrix} \beta_{11} & \beta_{12} & \cdots & \beta_{1K} \\ \beta_{21} & \beta_{22} & \cdots & \beta_{2K} \\ \vdots & \vdots & \ddots & \vdots \\ \beta_{M1} & \beta_{M2} & \cdots & \beta_{MK}. \end{pmatrix}$$

In particular, we have that model is *heteroskedastic* since

$$E[\epsilon_t \epsilon_t^T] = \text{diag}(\sigma_1^2, \sigma_2^2, \ldots, \sigma_M^2).$$

The model can also be written as a **time series regression** in the form

$$R_i = 1_T \alpha_i + F\beta_i + \epsilon_i \ \ i = 1, 2, \ldots, M.$$

In this case,

$$R_i = \begin{pmatrix} R_{i1} \\ R_{i2} \\ \vdots \\ R_{iT} \end{pmatrix}, \ 1_T = \begin{pmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{pmatrix}, \ \beta_i = \begin{pmatrix} \beta_{i1} \\ \beta_{i2} \\ \vdots \\ \beta_{iK} \end{pmatrix},$$

$$F = \begin{pmatrix} f_{11} & f_{21} & \cdots & f_{K1} \\ f_{12} & f_{22} & \cdots & f_{K2} \\ \vdots & \vdots & \ddots & \vdots \\ f_{K1} & f_{K2} & \cdots & f_{KT}. \end{pmatrix}, \ \epsilon_i = \begin{pmatrix} \epsilon_{i1} \\ \epsilon_{i2} \\ \vdots \\ \epsilon_{iT} \end{pmatrix},$$

and in this case $E[\epsilon_i \epsilon_i^T] = \sigma_i^2 I_T$ with I_T the $T \times T$ unit matrix, i.e., the error covariance is *homoskedastic*.

For the multivariate version there are at least 3 different ways to write it. The one we propose here is perhaps the most suitable for direct estimation:

$$R = X\Delta + E$$

where

$$R = \begin{pmatrix} R_1 \\ R_2 \\ \vdots \\ R_M \end{pmatrix}, \ Q = \begin{pmatrix} 1 & f_{11} & f_{21} & \cdots & f_{K1} \\ 1 & f_{12} & f_{22} & \cdots & f_{K2} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & f_{1T} & f_{2T} & \cdots & f_{KT} \end{pmatrix}, \ X = \begin{pmatrix} Q & \emptyset & \cdots & \emptyset \\ \emptyset & Q & \cdots & \emptyset \\ \vdots & \vdots & \ddots & \vdots \\ \emptyset & \emptyset & \cdots & Q \end{pmatrix}, \ E = \begin{pmatrix} \epsilon_1 \\ \epsilon_2 \\ \vdots \\ \epsilon_M \end{pmatrix}$$

where R_i and ϵ_i were defined above and $\emptyset$ is a$T \times (K+1)$ matrix of zeroes. Finally,

$$\Delta = (\alpha_1, \beta_{11}, \beta_{21}, \cdots, \beta_{K1}, \alpha_2, \beta_{12}, \beta_{22}, \cdots, \beta_{K2}, \cdots, \alpha_M, \beta_{1M}, \ \beta_{2M}, \cdots, \beta_{KM})^T$$

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and

$$\text{cov}(E) = \begin{pmatrix} \sigma_1^2 I_T & \emptyset & \cdots & \emptyset \\ \emptyset & \sigma_2^2 I_T & \cdots & \emptyset \\ \vdots & \vdots & \ddots & \vdots \\ \emptyset & \emptyset & \cdots & \sigma_M^2 I_T \end{pmatrix} \equiv D.$$

Note that $\text{cov}(E)$ is *homoskedastic, within each asset*, and the structure of the regression allows us to estimate the parameters Δ in an efficient manner using OLS for the individual time series regressions for each of the M assets.

However, computationally speaking, it might be convenient to use the multivariate regression setup. The matrix D is heteroskedastic, and in that case, we would need to use generalized or weighted least squares (GSL/WLS).

3. Factor Models and Covariance Structure

This section addresses the problem of estimation mentioned in the introduction.

In a traditional mean-variance optimization framework with M assets, we need to estimate many parameters: M returns,M variances, and $M(M-1)/2$ covariances: a total of $2M + M(M-1)/2$ parameters. If $M = 10$ that makes 65 parameters; if $M = 100$ that would be 5,150 parameters; for $M = 500$ it amounts to 125,750 parameters.

However, the problem does not lie so much in the number of parameters per se, but the fact that, using the traditional timeframes adopted for this type of analysis, the standard deviations of our estimates for the means are larger than the estimates themselves. That makes the estimates useless (**historical blur**) and could mislead a portfolio manager into a false sense of security about their optimized portfolio.

The only way around the issue would be to increase the number of data points, i.e., considering longer periods of time by going back 20, 30, 50, or more years, instead of the more traditional 2, 3, or 5. While this may not be enough to solve the problem, it is also not desirable since, over a very long period of time, we cannot assume that the parameters have not changed. When working with high-frequency data (as with Bitcoin and other cryptocurrencies) we would have more data, but that data could still contain a lot of noise and not allow for precise estimates.

Factor models can be of assistance. In fact, if we use the cross-section setup:

$$R_t = \alpha + Bf_t + \epsilon_t \;\; t = 1, 2, \ldots, T,$$

then under the assumptions of the multifactor model, the $M \times M$ covariance matrix of asset returns has the form

$$\text{cov}(R_t) = \Omega = B\Sigma_f B^T + \Sigma_\epsilon, \;\; \Sigma_\epsilon = \text{diag}(\sigma_1^2, \sigma_2^2, \ldots, \sigma_M^2).$$

This last result implies that

$$\text{var}(R_{it}) = \beta_i^T \Sigma_f \beta_i + \sigma_i^2, \;\; \text{cov}(R_{it}, R_{jt}) = \beta_i^T \Sigma_f \beta_j.$$

Notice that, given our assumption about the structure of the model and the stationarity, these estimates do not depend on t as long as the loading factors,B are time invariant (which we are assuming). Even accounting for the fact that one has to estimate B and Σ_f and the σ_i^2's, the amount of parameters is considerably smaller than using the direct approach.

Hence, the first contribution of factor models is that, under certain circumstances, they provide a way to alleviate the historical blur issue in estimating the parameters for a mean-variance optimization program.

Example (Porfolio factor model). Let $w = (w_1, \ldots, w_M)^T$ be the vector of weights that identify a porfolio of M assets. Then, the return on the portfolio is given by

$$R_{pt} = w^T R_t$$

If we use the (cross section) factor model for R_t, i.e.,

$$R_t = \alpha + Bf_t + \epsilon_t,$$

then

$$R_{pt} = w^T \alpha + w^T B f_t + w' \epsilon_t$$

which shows how the return on the portfolio is related to the factors chosen. In particular, $\beta_p^T = w^T B$ represents the loading/sensitivities of the portfolio with respect to the factors selected. In addition,

$$\mathbf{var}(R_{Pt}) = w^T B \Sigma_f B^T w + w^T \Sigma_\epsilon w.$$

4. Factors and Estimation of Coskweness and Cokurtosis Matrices.

If we have a model like

$$R_t = a + Bf_t + \epsilon_t,$$

where B as before is the matrix of factor loadings and F the matrix of factors, we have just seen how that allows us to estimate the covariance matrix using fewer parameters. What happens to the coskewness and cokurtosis matrices? Well, there are formulas linking the matrices B and F to the coskewness and cokurtosis matrices. Using the same symbols we used in Lesson 2, these are:

$$M_3 = BM_{3,f}(B^T \otimes B^T) + \Omega$$

$$M_4 = BM_{4,f}(B^T \otimes B^T \otimes B^T) + \Gamma.$$

Here $M_{3,f}$ and $M_{4,f}$ are the coskweness and cokurtosis matrices for the factors.Ω and Γ here are special matrices. Specifically,Ω is an $N \times N^2$ matrix of zeroes, except for the i, j elements where $j = (i-1)N + h$ (recall that for coskewness we need three indices) corresponding to the expected third moment of the terms ϵ_t. Γ is more complicated, unfortunately. The definitions and the proofs of these decompositions are provided in a paper by Boudt et al. listed among the references, but which is not open access, unfortunately. A link to a paper, which is freely available, is provided in the references (albeit the code therein is in R and not in Python).

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5. Macroeconomic Factor Models

Arguably, the most famous example of a macroeconomic factor model is the **single factor model** introduced by Sharpe in 1970. This model is:

$$R_{it} = \alpha_i + \beta_i R_{Mt} + \epsilon_{it}, \;\; i = 1, 2, \ldots, N; t = 1, 2, \ldots, T$$

where R_{Mt} is the return of the market index in excess of the risk-free rate and it is called the **market risk factor.**

*(**Note:** We change the index M to N to avoid confusion with the market risk factor).*

In this case,R_{it} are the returns of asset i in excess of the risk-free rate as well. Then, for instance, we could use this model to test if the CAPM hold ($\alpha_i = 0$ for each i). In this setting we have a single factor i.e.$f_{1t} = R_{Mt}$. It is easy to see that in this particular model:

$$\sigma_M^2 = \mathbf{var}(R_{Mt}), \;\; \beta^T = (\beta_1, \ldots, \beta_N)^T, \;\; \mathbf{cov}(R_t) = \sigma_M^2 \beta\beta^T + \Psi$$

with $\Psi = \mathbf{diag}(\sigma_1^2, \sigma_2^2, \ldots, \sigma_N^2)$.
Regarding the estimation of this model, if we use the time series regression models for each asset i, these models satisfy the homoskedasticity assumption, and thus, OLS are the best linear unbiased estimates. Hence, let $\widehat{\alpha}_i, \widehat{\beta}_i, i = 1, 2, \ldots, N$ be the OLS estimates.

Then, we can estimate the residuals as

$$\hat{\epsilon}_{it} = R_{it} - \widehat{\alpha}_i - R_{Mt}\widehat{\beta}_i$$

and

$$\widehat{\sigma}_i^2 = \frac{1}{T-2}\sum_{t=1}^{T} \hat{\epsilon}_{it}^2, \;\; i = 1, 2, \ldots, N.$$

This allows us to obtain $\widehat{\Psi} = \mathbf{diag}(\widehat{\sigma}_1^2, \ldots, \widehat{\sigma}_N^2)$. Finally,

$$\widehat{\sigma}_M^2 = \frac{1}{T-1}\sum_{t=1}^{T}(R_{Mt} - \bar{R}_M)^2$$

and the unconditional covariance for R_t is then given by

$$\widehat{\mathbf{cov}(R_t)} = \widehat{\sigma}_M^2 \widehat{\beta}\widehat{\beta}^T + \widehat{\Psi}.$$

The extension to the multivariate case is straightforward as one can estimate M time series regressions:

$$\begin{pmatrix} \widehat{\alpha}_i \\ \widehat{\beta}_{1i} \\ \widehat{\beta}_{2i} \\ \vdots \\ \widehat{\beta}_{Ki} \end{pmatrix} = (Q^T Q)^{-1} Q R_i \;\; i = 1, 2, \ldots, M,$$

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and where the matrix q was defined earlier.

The estimated residuals can be obtained as usual

$$\hat{\epsilon}_{it} = R_{it} - \widehat{\alpha}_i - Q\widehat{\beta}_i$$

so that

$$\widehat{\sigma}_i^2 = \frac{1}{T-K-1}\sum_{t=1}^{T}\hat{\epsilon}_{it}^2.$$

Finally,

$$\widehat{\Sigma}_f = \frac{1}{T-1}\sum_{t=1}^{T}(f_t - \bar{f})(f_t - \bar{f})^T \quad \text{and}$$

$$\widehat{\Sigma} = \widehat{B}\widehat{\Sigma}_f\widehat{B}^T + \widehat{\Psi}.$$

None of these regressions present issues since the factors are observable.

6. Fundamental Factors

The goal of fundamental factor models is to identify features that are relevant to forecasting potential return and risk for securities. Some of these factors are microeconomic (industry, earnings growth, price-to-book value, etc). Fundamental models decompose risk using a few easily understood elements. Intuitively appealing, fundamental models are ideal for performance attribution and portfolio risk decomposition. They are also supported by decades of academic research in asset pricing theory. However, since fundamental models are defined in terms of known factors, they can become less accurate when new, and until then unknown, factors emerge as relevant.

Arguably, one of the most famous fundamental models is the **Fama and French Three Factor Model**. This model uses three factors:

$$R_{it} - r_{ft} = \alpha_{it} + \beta_1(R_{Mt} - r_{ft}) + \beta_2 SMB_t + \beta_3 HML_t + \epsilon_{it}$$

where:

- R_{it} is the return of asset i at time t;
- r_{ft} is the risk-free rate of return at time t;
- R_{Mt} is the return on the market portfolio;
- $R_{it} - r_{ft}$ and $R_{Mt} - r_{ft}$ are excess returns on asset i and the market portfolio, respectively;
- $R_{Mt} - r_{ft}$ is the market premium (momentum factor);
- SMB_t is the size premium (small minus big). This time series represents the historical excess returns of small companies over large cap companies (*size factor*);
- HML_t is the value premium (high minus low). This time series contains the historical excess returns of value stocks vs. growth stocks. A stock is a value stock if it has a high book-to-price ratio, and it is a growth stock if it has a low book-to-price ratio (*value factor*) instead.

The values for SMB and HML time series and explanations about the methodology are found on Kenneth French's website at Dartmouth University (Kenneth French's datasets).

Example. As an application, we can estimate a 3-factor model for Apple stock during the period January 2017 to February 2022.

The OLS regression is shown below. We notice that:

- the α is statistically not significant, which is what we should expect if the asset pricing theory (APT) model holds.
- The coefficient of the market risk $\beta_1 = 1.259$ is like a beta (it would be if we had estimated the model without the other regressors) and shows that the market returns affect the returns of Apple stock as well (momentum);
- $\beta_2 = -0.4903$ tells us that the size factor has a negative impact, i.e., Apple, being a large cap company, is actually less risky and thus gets a discount in terms of returns. However,β_2 is not statistically significant at the 90%-confidence level;
- $\beta_3 = -0.5313$ is statistically significant and the value factor has a negative impact. In other words, Apple qualifies like a growth stock rather than a value stock.

Figure 1: Estimated 3-Factor Model for Apple during the Period January 2017 to February 2022

OLS Regression Results

Dep. Variable:	excess_rtn	R-squared:	0.477
Model:	OLS	Adj. R-squared:	0.449
Method:	Least Squares	F-statistic:	17.04
Date:	Wed, 02 Mar 2022	Prob (F-statistic):	5.50e-08
Time:	11:06:35	Log-Likelihood:	82.465
No. Observations:	60	AIC:	-156.9
Df Residuals:	56	BIC:	-148.6
Df Model:	3		
Covariance Type:	nonrobust		

	coef	std err	t	P>|t|	[0.025	0.975]
Intercept	0.0140	0.009	1.622	0.110	-0.003	0.031
mkt	1.2590	0.185	6.812	0.000	0.889	1.629
smb	-0.4903	0.316	-1.550	0.127	-1.124	0.143
hml	-0.5313	0.206	-2.580	0.013	-0.944	-0.119

Omnibus:	6.222	Durbin-Watson:	2.204
Prob(Omnibus):	0.045	Jarque-Bera (JB):	6.386
Skew:	-0.443	Prob(JB):	0.0410
Kurtosis:	4.330	Cond. No.	39.7

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

Fama and French also have models with 4 or 5 factors. Chen et al. published a model with 5 factors as well.

- industrial production,
- changes in the risk premium,
- twists in the yield curve,
- measures of unanticipated inflation,
- changes in inflation during high volatile periods for these variables.

This model, unlike Fama and French, contains only observed macro-economic factors. Fung and Hsieh proposed a risk-factor model with 8 factors, which is applied to hedge funds. Barra Inc.'s Global Equity Model (GM3) uses 34 industry factors based on GICS. Axioma provides world-wide

equity factor risk models. Other factor models are supplied by Morningstar and Bloomberg, etc. These indices are mostly not free as it is a lot of work to update the time series, rebalance them, etc.

The Barra model incorporates two types of factors: style and industry. Barra style factors include momentum, size, earning yield, residual volatility, growth, dividend yield, book-to-price, leverage, liquidity, etc.

The industry factors are not necessarily what one thinks, at least not completely, as defining an industry is somewhat arbitrary. We will discuss this in the next section. However, their model looks like this:

$$R_{it} = \alpha_i f_{ct} + \sum_{k=1}^{K} \beta_{ik} f_{kt}^{(I)} + \sum_{h=1}^{H} \beta_{ih} f_{ht}^{(S)} + \epsilon_{it}, \;\; i = 1, 2, \ldots, N$$

where the $f_k^{(I)}$ factors are industry sectors and the $f_h^{(s)}$ factors are style factors. In this case, the loading matrix vector β_i is the exposure of asset i to the industry factors. If stock i is in industry g, then $\beta_{ig} = 1$ and $\beta_{ig} = 0$ when $i \neq g$. f_c is a country factor, in which case every stock has an exposure of 1 to its country factor.

7. Statistical Factors

The main difference among statistical/latent factors and the previous two groups is that statistical factors are not observable. And it is a big difference!

To understand the big change, we can start with a relatively easy situation like an industry factor model in which we assume that there are H industries. We could choose H as large as the number of stocks, but that would defeat the purpose, or we could choose H as a number of, say, five industries. Now, five may not be enough, but the emphasis here is on the nuances of the methodology and not its accuracy. Let's say that we have N risky assets and $H < N$ industries. Then, an industry factor model looks like this:

$$R_{it} = \gamma_{i1} f_{1t} + \gamma_{i2} f_{2t} + \ldots + \gamma_{iH} f_{Ht} + \epsilon_{it}, \;\; i = 1, 2, \ldots, N; t = 1, 2, \ldots T$$

where we still have $\mathbf{var}(\epsilon_{it}) = \sigma_i^2$ for $i = 1, 2, \ldots, N$, $\mathbf{cov}(f_{ht}, \epsilon_{it}) = 0$ for $h = 1, \ldots, H$ and $j = 1, \ldots N$, and $\mathbf{cov}(\epsilon_{it}, \epsilon_{is}) = 0$ for $s \neq t$. However, now we have additional assumptions:

$$\gamma_{ih} = \begin{cases} 1 & \textbf{if asset i is in industry h;} \\ 0 & \textbf{otherwise.} \end{cases}$$

Notice the key difference: _We are assuming that the γ_{ik} are the known parameters, while the factors are the unknown_. Hence, in this type of model, we will estimate the factors and not the γ's. Also, if we assume that there are N_h assets in industry h, $h = 1, 2, \ldots, H$, then $\sum_{h=1}^{H} N_h = N$.

In addition, since the industries in our definition are mutually exclusive, we also have $\gamma_i^T \gamma_h = N_h$ if $j = h$ or zero otherwise. Unlike before, in this case, we do not know the matrix Σ_f since we do not observe the factors. Nevertheless, we need the time series of factors in order to estimate the residuals and the covariance $D = \mathbf{diag}(\sigma_1^2, \sigma_2^2, \ldots, \sigma_n^2)$ which will in turn be used to compute the WLS estimates. As we are using cross sections, we need to use GLS/WLS because OLS provides

unbiased estimates but not efficient ones due to the fact that there is cross-sectional heteroskedasticity in the returns of the assets, as we learned in the setup section.

In this case, a multi-step estimation process known as **FGLS** (**feasible generalized least squares**) is required. It works like this:

Step 1: Estimate the factors using OLS which yields:

$$\hat{f}_{ht} = \frac{1}{N_h} \sum_{j=1}^{N_h} R_{jt}^{(h)} \;\; h = 1, 2, \ldots, H$$

where the notation $R_{jt}^{(h)}$ is a reminder that we are considering only the returns at time t of those stocks that belong to industry h. If we used a matrix notation, we could write the results as follows (this is just a more compact multivariate regression notation):

$$\widehat{f}_t^{OLS} = (\Gamma^T \Gamma)^{-1} \Gamma^T R_t.$$

Step 2: Estimate the factor covariance matrix. This is done from the estimates for the factors we found in Step 1. Hence

$$\widehat{\Sigma}_f^{OLS} = \frac{1}{T-1} \sum_{t=1}^{T} (\widehat{f}_t^{OLS} - \bar{f}^{OLS})(\widehat{f}_t^{OLS} - \bar{f}^{OLS})^T, \;\; \bar{f}^{OLS} = \frac{1}{T} \sum_{t=1}^{T} \widehat{f}_t^{OLS}.$$

Step 3: Once the estimates 's are available, we can compute the estimated residuals $\hat{\epsilon}_{ht}$'s. In turn, from these estimated residuals we can compute $\widehat{D}^{OLS}$ using

$$\widehat{\sigma}_i^{2,OLS} = \frac{1}{T-1} \sum_{t=1}^{T} (\hat{\epsilon}_{it}^{OLS} - \bar{\epsilon}_i^{OLS})^2 \;\; i = 1, 2, \ldots, N.$$

Step 4: This step is simple addition:

$$\widehat{\Sigma}^{OLS} = B \widehat{\Sigma}_f^{OLS} B^T + \widehat{D}^{OLS}.$$

Step 5: We now use GLS/WLS for the final step. Thus:

$$\widehat{f}_t^{GLS} = \left(B^T (\widehat{D}^{OLS})^{-1} B\right)^{-1} B^T (\widehat{D}^{OLS})^{-1} R_t, \;\; t = 1, 2, \ldots, T;$$

$$\widehat{\Sigma}_f^{GLS} = \frac{1}{T-1} \sum_{t=1}^{T} (\widehat{f}_t^{GLS} - \bar{f}^{GLS})(\widehat{f}_t^{GLS} - \bar{f}^{GLS})^T, \;\; \bar{f}^{GLS} = \frac{1}{T} \sum_{t=1}^{T} \widehat{f}_t^{GLS};$$

$$\widehat{\sigma}_i^{2,GLS} = \frac{1}{T-1} \sum_{t=1}^{T} (\hat{\epsilon}_{it}^{GLS} - \bar{\epsilon}_i^{GLS})^2 \;\; i = 1, 2, \ldots, N;$$

$$\widehat{\Sigma}^{GLS} = B \widehat{\Sigma}_f^{GLS} B^T + \widehat{D}^{GLS}.$$

For the industry factors we considered above, we could not observe the factors. However, because of the particular structure of the loadings, we knew the matrix B. The most general case is when the common factors are hidden and the loadings are not known.

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We do know the $N \times N$ covariance matrix of returns for our risky assets Σ, though. In this case, the factors must be extracted from Σ, which contains our "knowledge" about the problem. This can be done using either principal component analysis (PCA) or factor analysis. The steps using PCA are the following:

Step 1: Obtain $\widehat{\Sigma}$, the sample covariance. We have done this several times already.

Step 2: The matrix $\widehat{\Sigma}$ admits the following (spectral) representation $\widehat{\Sigma}_f = \Gamma \Lambda \Gamma^T$ where $\Lambda = \mathbf{diag}(\lambda_1, \lambda_2, \ldots, \lambda_K)$ with $\lambda_1 \geq \lambda_2 \geq \ldots \geq \lambda_K > 0$ the eigenvalues of $\widehat{\Sigma}_f$, and Γ the eigenvectors associated to those eigenvalues and such that $\Gamma\Gamma^T = \Gamma^T\Gamma = I_N$.

Step 3: Obtain initial estimates:

$$\tilde{\alpha}_0 = \bar{R} = \frac{1}{T} R \iota^T.$$

$$\tilde{B}_0 = \Gamma^{(k)} (\Lambda^{(k)})^{1/2}$$

where $\Gamma^{(k)}$ contains the first k columns of Γ and $\Lambda^{(k)}$ the first k columns of Λ.

$$\tilde{\Psi}_0 = \mathbf{diag}(\widehat{\Sigma}) - \mathbf{diag}(\tilde{B}_0 \tilde{B}_0^T).$$

$$\tilde{\Sigma}_0 = \tilde{B}_0 \tilde{B}_0^T + \tilde{\Psi}_0,$$

Step 4: Recompute the sample covariance matrix as $\widehat{\Sigma}^\star = \widehat{\Sigma} - \tilde{\Psi}_0$ Then, recompute the decomposition $\widehat{\Sigma}^\star = \tilde{\Gamma} \tilde{\Lambda} \tilde{\Gamma}^T$.

Step 5: Repeat step 3 using the new elements to generate $\tilde{B}_1$ and $\tilde{\Psi}_1$. Then, use these to generate $\tilde{\Sigma}_1 = \tilde{B}_1 \tilde{B}_1^T + \tilde{\Psi}_1$.

Step 6: Repeat step 5 several times to generate a sequence of estimates $\tilde{B}_u, \tilde{\Psi}_u, \tilde{\Sigma}_u$ for $u = 2, 3, \ldots$ and stop when the changes in $\tilde{\Psi}_u$ and $\tilde{\Psi}_{u+1}$ are negligible. These are the final estimates.

When we use a spectral decomposition, the eigenvectors matrix Γ contains orthogonal vectors. Each of this vectors is one component in the matrix of sample principle components because the matrix $\widehat{\Sigma}$ defined by $\Gamma \Lambda \Gamma^T$ is the covariance matrix of the sample principal component $\widehat{P} = \Gamma R$. Now, $\widehat{P}$ is an $M \times T$ matrix, and we can choose as many principal components as necessary "to do a good job," not necessarily all of them.

One question left to answer is how many factors should one use? If we consider again the spectral decomposition of Σ or $\widehat{\Sigma}$, i.e.

$$\widehat{\Sigma} = \Gamma \Lambda \Gamma^T \;\; \Gamma^{-1} = \Gamma^T,$$

the matrix Γ consists of orthonormal eigenvectors. For principal component analysis, one would take the diagonal of Λ and sort its elements into descending order. The same sorting procedure must be executed on the columns of Γ, of course. Then, the fraction of the variance explained by each principal component is given by the ratios:

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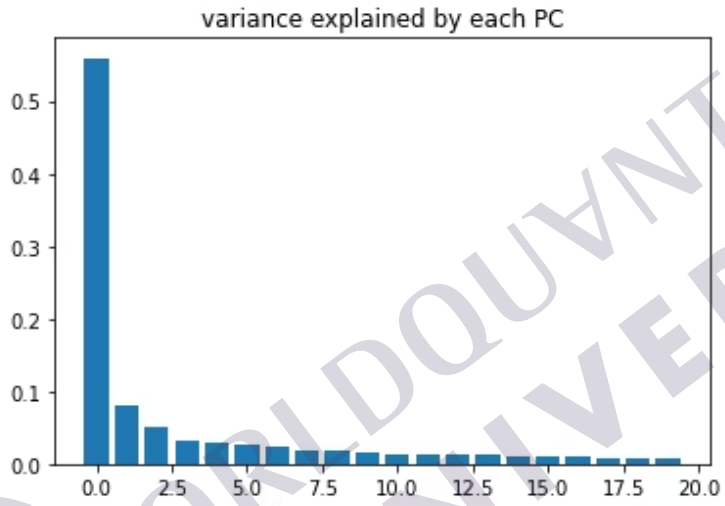
$$\frac{\lambda_i}{\sum_{i=1}^{N}\lambda_i}, \quad \text{and} \quad \frac{\sum_{i=1}^{I}\lambda_i}{\sum_{i=1}^{N}\lambda_i},$$

which represent the percentage of the total variance explained by the i-th principal component of factor returns and the ratio of the total variance explained by the first I principal components, respectively.

This can help you decide how many factors to use.

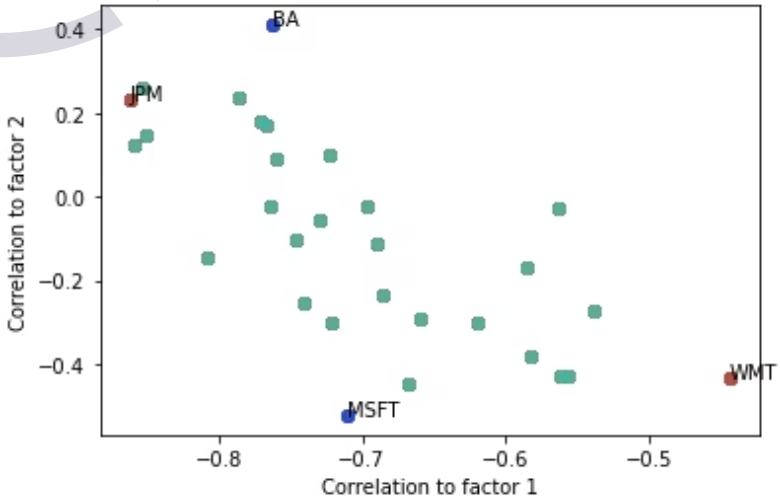
Example (Extracting principal components for the Dow Jones Industrial Average Index). If we considered the DJ Index 30 companies during the period January 2, 2020–March 4, 2022, we find that the first two principal components explain about 2/3 of the volatility as shown in figure 2.

Figure 2: Twenty Principal Components for the Dow Jones Industrial Index over the Period Jan. 2, 2020 - Mar. 4, 2022_



The next figure show the correlations of each of the 30 components of the Dow Jones Index with factor 1 and factor 2. The dots in red and blue denote the maximum and minimum correlations with the first and second factors, respectively. Principal components, when used as factors, are not easy to explain as recognizable macroeconomic or fundamental factors.

Figure 3: Correlations of Returns of Dow Jones Industrial Average 30 Individual Assets with the First and Second Principal Components Considered as Factors



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8. Factor-Mimicking Portfolio (FMP)

Assume that we have identified a factor that is relevant to us. However, the factor may not be observable in the market. How can we obtain the excess returns for it when it is not observable?

The answer is that we can create a portfolio that behaves like this factor using a group of observable ones for which returns or excess returns are available.

The idea is this: We have N risky assets for which the $N \times N$ covariance matrix of returns,Ω, is known, we also have K factors. Then, if w_k is an $N \times 1$ vector of weights, we solve the following variance minimization program:

$$\min_{w_k} \frac{1}{2} w_k^T \Omega w^T$$
$$\text{subject to } B^T w_k = \delta_k.$$

It is important to understand this part. The objective function is clear. The constraints employ B, the loading matrix we have used before. The constraints, written explicitly, look like this:

$$B^T w_k = \begin{pmatrix} \beta_{11} & \beta_{21} & \cdots & \beta_{N1} \\ \beta_{12} & \beta_{22} & \cdots & \beta_{N2} \\ \vdots & \vdots & \ddots & \vdots \\ \beta_{1K} & \beta_{2K} & \cdots & \beta_{NK} \end{pmatrix} \cdot \begin{pmatrix} w_{k1} \\ w_{k2} \\ \vdots \\ w_{kN} \end{pmatrix} = \begin{pmatrix} \sum_{i=1}^N \beta_{i1} w_{ki} \\ \sum_{i=1}^N \beta_{i2} w_{ki} \\ \vdots \\ \sum_{i=1}^N \beta_{iK} w_{ki} \end{pmatrix} \equiv \begin{pmatrix} \delta_{k1} \\ \delta_{k2} \\ \vdots \\ \delta_{kK} \end{pmatrix} \equiv \delta_k.$$

Hence,δ_k is a$K \times 1$ vector for each value of $k = 1, 2, \ldots, K$. Now,δ_{k1} represents the loading of factor 1 for the portfolio,δ_{k2} the loading of factor 2, and so on. In particular,δ_{kk} represents the loading of factor k for the portfolio. These δ_{kh} elements are our choice. We can solve the optimization problem as we did before when dealing with Markowitz mean-variance optimization—just keep in mind that here we technically have K constraints. We will not go over the entire derivation via calculus, but the result is

$$w_k^\star = \Omega^{-1} B (B^T \Omega^{-1} B)^{-1} \delta_k.$$

The factor-mimicking portfolio weight matrix (for all K factors) is given by:

$$W_K^\star = \Omega^{-1} B (B^T \Omega^{-1} B)^{-1} \Delta_K.$$

where Δ_K is the $K \times K$ matrix $(\delta_1, \delta_2, \ldots, \delta_K)$.

Now, if we want the portfolio to mimic factor k, it seems natural that we must have $\delta_{kk} = 1$. For δ_{kh} when $h \neq k$, we do not need to assign them; nevertheless, if we set $\delta_{kh} = 0$ for $h \neq k$, we obtain a **pure factor-mimicking portfolio for factor k.**

Example: We can look at what happens if we use the formula just derived for the FMP weights under certain situations:

- $\Omega = \sigma^2 I_N$ (assets are uncorrelated and have constant variance) and $\Delta_K = I_K$, the $K \times K$ identity matrix. In this case, we find $W_K^\star = B(B^T B)^{-1}$.
- $\Omega = D \equiv \text{diag}(\sigma_1^2, \sigma_2^2, \ldots, \sigma_N^2)$, that is, the assets are uncorrelated but are allowed to have different variances and $\Delta_K = I_K$. In this case, we get $W_K^\star = D^{-1} B (B^T D^{-1} B)^{-1}$.

- In this case, we assume that Ω is unconstrained, but $\Delta_K = I_K$. Then, $W_K^\star = \Omega^1 B(B^T\Omega^{-1}B)^{-1}$.

Example. (Factors and loadings are not unique). Consider the model
$R_t = \alpha + Bf_t + \epsilon_t, \;\; t = 1, 2, \ldots, T$
(the cross-section regression notation) as done in the setup section. Now, for any $K \times K$ matrix that is invertible we can define
$f_t^\star = Hf_t$ and $B^\star = BH^{-1}$. We note the following:

$$R_t = \alpha + Bf_t + \epsilon_t = R_t = \alpha + BH^{-1}Hf_t + \epsilon_t = \alpha + B^\star f_t^\star + \epsilon_t.$$

Furthermore, we have

$$\mu_f^\star = H\mu_f, \;\; \Sigma_f^\star = \text{cov}(f_t^\star) = \text{cov}(Hf_t) = H\Sigma_f H^T.$$

The matrix H is our choice. If Σ_f is positive definite, it admits a (spectral) decomposition
$\Sigma_f = \Gamma\Lambda\Gamma^T$
where $\Lambda = \text{diag}(\lambda_1, \lambda_2, \ldots, \lambda_K)$ with $\lambda_1 \geq \lambda_2 \geq \ldots \geq \lambda_K > 0$ the eigenvalues of Σ_f, and Γ the eigenvectors associated with the eigenvalues. Then, if we let $H = \Gamma\Lambda^{-1/2}$, we obtain
$\Sigma_f^\star = I_K$.

Since factors and loadings are not unique, this creates some issues for their interpretation. Using the matrix H as done above is equivalent to using **orthogonal transformations**, which amount to **rotation** of the coordinates. Some rotation may be easier to interpret than others.

9. Returns-Based Style Analysis

When performing a returns-based style analysis, the goal is to identify the style of a fund through a regression of the fund's returns against a certain number of passive style indices. There are many indices available. For instance, one could check the MSCI website.

In certain applications, say that we want to understand how a portfolio manager achieves results. We can use a regression over a pre-specified period of time and see where the manager's exposure lies. This can inform us about the manager's style in investing. Comparing the regression coefficients over different periods of time, we can detect if a change in style has occurred.

Example: Let's say that we have the returns for the manager's portfolio over a period of T months, $R_{pt}, t = 1, 2, \ldots, T$. For the same period, we also have the returns for 4 indices: small-cap growth (SCG), large-cap growth (LCG), small-cap value (SCV), and large-cap value (LCV). Then, we estimate the following linear regression model:

$$R_{pt} = \alpha + \beta SCG_t + \gamma LCG_t + \delta SCV_t + \theta LCV_t + \epsilon_t, \;\; t = 1, 2, \ldots, T$$

with the constraints that $\beta + \gamma + \delta + \theta = 1$ (constrained linear regression model).
The estimated coefficients are found to be:

$$\hat{\beta} = 0.02, \hat{\gamma} = 0.05, \hat{\delta} = 0.15, \hat{\theta} = 0.78.$$

What can we say about the manager's investment style?

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- The portfolio has very little exposure to growth stocks ($\hat{\beta}$ and $\hat{\gamma}$ are very small);
- As a consequence, the manager invests mostly in value stocks. In addition, since $\hat{\theta}$ is much larger than $\hat{\delta}$, we can also state that the manager is primarily investing in large-cap value stocks.

Assume that now, it is five years later, and using new data about the manager's portfolio returns and the indices, we estimate the same regression model and this time

$$\hat{\beta} = 0.15, \hat{\gamma} = 0.30, \hat{\delta} = 0.20, \hat{\theta} = 0.35.$$

In this case, the shift in the manager's style is quite evident.

- The portfolio is now nearly split between growth and value assets;
- While large cap are still preferred, there is definitely less emphasis on them vs. the previous period.

10. Are Factors Helping in Building Better Portfolios?

We have seen enough of factors to be aware that factor is just a label for a portfolio of investable securities. In principle, the lower correlations among factors (even negative at times) vs. those among assets, seem to imply that factors offer the potential for stronger diversification. Asset classes are often positively correlated for a comparison. Nevertheless, it has been pointed out that the low correlations one observes among factors are the consequence of the fact that *replicating portfolios* contain short positions. If we allowed the same amount of short positions in portfolios built from asset classes, then the difference in the computed efficient frontier vanishes. An interesting experiment would be to compare a portfolio built with a certain number of assets to a portfolio built using, say, the first two principal components for the covariance matrix of those same assets (principal components, by construction, are uncorrelated) and allowing for the same amount of short positions in the portfolio of assets. Empirical evidence suggests that the results are very similar. Hence, some may call this supposed superiority of factors a fallacy in asset allocation.

11. Conclusion

This lesson completes the treatment of classical portfolio theory. In this lesson, we looked at three types of factors (macroeconomic, fundamental, and statistical). And, using factor models, we were able to reduce the number of parameters to estimate to perform mean-variance optimization. This holds true for any kind of optimization problem and not just Markowitz's model as long as there are parameters to estimate. However, factor models can allow us to identify a few relevant common sources of risk and can be used to create portfolios. With statistical factors, we learned to deal with factors that exist but are not directly observable. Finally, using specific indices (that to a large extent could be a result of a planned exposure to specific sources of risk/factors), we can identify portfolio managers' investment styles and assess if their styles have changed over certain periods of time. As such, factor models can become a tool of their own.

The next lesson introduces the well-known Black-Litterman model, which allows investors to introduce their feeling for markets and assets.

References

- Barra. "Barra Global Equity Model (GEM3)." MSCI. https://www.msci.com/documents/10199/242721/Barra_Global_Equity_Model_GEM3.pdf
- Boudt, Kris et al. "Higher Order Comoments of Multifactor Models and Asset Allocation." Finance Research Letters, vol. 13, 2015, pp. 225–233.
- Chen, Nai-Fu et al. "Economic Forces and the Stock Market." The Journal of Business, vol. 59, no. 3, 1986, pp. 383–403.
- Cornilly, Dries, and Brian G. Peterson. "PerformanceAnalytics: Estimation of Higher Order Moments." 2018. https://cran.r-project.org/web/packages/PerformanceAnalytics/vignettes/EstimationComoments.pdf
- French, Kenneth R. "Description of Fama/French Factors." https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/Data_Library/f-f_factors.html
- MIT Open Courseware. Factor Analysis Model. 2013. https://ocw.mit.edu/courses/mathematics/18-s096-topics-in-mathematics-with-applications-in-finance-fall-2013/lecture-notes/MIT18_S096F13_lecnote15.pdf

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